A Stratified Multi-Rater Evaluation of LLM-Based Virtual Standardized Patients with a Deployed Data-Generation Platform

Yongjin Yi*
Internal Medicine, College of Medicine,
Dankook University
yongjin.neph@gmail.com

Jinyong Park*
Department of Medical Device Development,
Seoul National University
nwirandx@gmail.com

Sejoong Kim
Internal Medicine, College of Medicine,
Seoul National University
sejoong2@snu.ac.kr

Abstract

The Objective Structured Clinical Examination (OSCE) is widely used to assess clinical communication skills in medical education, requiring trainees to take histories and reason clinically with standardized patients (SPs). Large language model (LLM)-based virtual standardized patients (VSPs) offer a practical alternative to human SPs. These are typically built using persona-injected prompting, which casts a model into a clinical role via a single system prompt. However, existing evaluations of such VSPs often rely on a single backbone LLM, small human rater panels, or LLM-as-a-judge scoring without an utterance-aligned human-expert reference. Consequently, they are limited in their ability to identify the specific shortcomings of individual models. In this study, we evaluated four LLMs (GPT-4o, Claude 3.5 Sonnet, Llama-3.1 70B, and HyperCLOVA X), spanning commercial, open-source, and Korean-developed models, against a human-expert reference across three Korean OSCE cases (hematuria, fever in pregnancy, and chronic cough). In a double-blind, fully crossed design, 17 senior medical-student raters scored every model utterance on three sentence-level and six encounter-level Likert metrics, yielding 7,480 data points. We analyzed the scores using a stratified linear mixed-effects model based on an 11-category history-taking and 3-level linguistic-form taxonomy. At the encounter level, three of the four LLMs fell 0.08 to 0.47 Likert points below the human-expert reference on Comprehensibility, Consistency and Hallucination-freeness. The difference was not statistically significant for Hallucination-freeness in Claude 3.5 Sonnet and GPT-4o, Consistency in GPT-4o, and Engagement in HyperCLOVA X. Llama-3.1 70B fell significantly below the reference on all six metrics. Furthermore, we introduce a web-based OSCE platform that supports virtual practice across diverse clinical cases. This platform continuously collects student–VSP dialogues under research and privacy consent; the resulting dataset is available at https://huggingface.co/datasets/nwirandx/Korean-OSCE-history-taking-dialogues.

*Equal contribution.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

1 Introduction

The Objective Structured Clinical Examination (OSCE) has been the gold standard for assessing clinical skills (history taking, communication, clinical reasoning, and patient education) in medical education for decades [1]. National licensure programs in Taiwan, Canada, Germany, Spain, and the Republic of Korea have incorporated OSCE-format clinical examinations [2, 3], and OSCE-style training has been shown to shift student preparation toward integrated clinical competencies [4]. Despite this established role, the OSCE administration imposes substantial costs and accessibility barriers. Standardized patient (SP) programs require sustained investment in recruitment, training, examiner panels, and dedicated facilities. These costs, together with concerns about discriminative validity given consistently high pass rates, contributed to the 2021 retirement of the United States Medical Licensing Examination (USMLE) Step 2 Clinical Skills exam [5]. In the same year, the Medical Council of Canada terminated the MCCQE Part II OSCE for similar reasons of cost-effectiveness and redundancy with residency-based competency assessments [6, 7]. Although the educational rationale for the OSCE remains intact [8], the discontinuation of national clinical-skills examinations has necessitated cost-effective alternatives that preserve the formative function of OSCE-based training.

Large language models (LLMs) have rapidly emerged as a candidate substrate for such alternatives. Frontier models encode sufficient clinical knowledge to surpass the USMLE passing threshold [9] and approach expert performance on USMLE-style soft-skills items covering ethics, communication, and empathy [10]. Building on this capacity, several groups have implemented LLM-based virtual standardized patients (VSPs) via *persona injection*, a system-prompt-based approach in which the model is instructed to play a clinical role and answer utterance by utterance from within it. Despite its technical simplicity, this approach is plausible enough to recover several pedagogically important behaviors of human SPs, opening the prospect of deployable, low-marginal-cost training environments for clinical communication.

Research on LLM-based VSPs has progressed along three axes: feasibility and usability in history-taking practice [11, 12], educational efficacy demonstrated through randomized controlled trials of LLM-simulated encounters [13, 14], and agent-level dialogic capabilities approaching those of primary-care physicians [15]. Two recent syntheses [16, 17] conclude that the field is maturing rapidly, but that current evidence is heterogeneous in evaluation methodology and concentrated in English-language, single-scenario studies. Despite this growth, several gaps remain in the evaluation of LLM-based VSPs. Most studies have evaluated a single LLM on a single scenario, often with small rater panels and without formal inter-rater reliability analysis, so fully-crossed multi-rater comparisons against a human-expert baseline on identical clinical tasks are scarce. Existing reports also rely largely on aggregate Likert means or single composite scores, which obscure where a model fails: a single rank cannot distinguish a model that mishandles greetings from one that misinterprets clarifying questions about symptom character. In addition, almost all published LLM-VSP evaluations have been conducted in English, although diagnostic dialogue is constructed across many languages and clinical traditions; whether the same persona-injection prompt yields comparable behavior when the question is phrased in a language whose yes/no semantics differ from English is largely untested. The deployable utility of LLM-based VSPs relative to human SPs on specific dimensions of clinical communication remains an open question.

We address these gaps with a stratified, multi-rater human-evaluation framework for LLM-based VSPs, accompanied by a deployed training platform that enables the same evaluation to extend beyond a static set of prompts. Our work makes the following contributions.

1. We report a fully-crossed multi-rater evaluation against a human-expert reference. We evaluate four frontier LLMs (GPT-4o, Claude 3.5 Sonnet, Llama-3.1 70B, HyperCLOVA X) alongside a human medical-education expert serving as the SP-equivalent reference, across three OSCE cases. Seventeen final-year medical-student raters produced 7,480 sentence- and encounter-level ratings under formal inter-rater reliability analysis.
2. Every medical student input prompt is tagged with an 11-category history-taking taxonomy grounded in AMEE Guide No. 81 [1] and with a three-level linguistic-form label (affirmative, negative, open-ended). The taxonomy surfaces failure modes that aggregate Likert scores would otherwise hide, including a Korean negative-form penalty on Fluency.

3. We deploy a real-time persona-injected VSP web application on which medical students conduct free-form OSCE-style encounters, and we show that the same taxonomy applies to dialogue collected on the platform. The platform-collected corpus is released under informed consent on a public Hugging Face repository with Croissant Core and Responsible-AI metadata.

2 Related Work

LLM-based virtual standardized patients. LLM-based virtual standardized patients have most often been built using a single declarative system prompt, called persona injection, that fixes the patient's identity, clinical state, history, and conversational style, and constrains turn-by-turn answers to remain in role [18]. Within this paradigm, several representative systems anchor the field. Holderried et al. [11] developed a GPT-powered simulator for abdominal pain history-taking and validated it through mixed-methods qualitative interviews with German medical students. Elhilali et al. [12] developed a communication-skills simulator with user-centered usability as the principal outcome. Brügge et al. [13] conducted a randomized controlled trial in which structured feedback following LLM-simulated encounters improved medical students' clinical decision-making, as assessed against expert rubrics. Lavigne et al. [14] reported the first deployment-scale RCT (n = 247) of an LLM-based OSCE-training platform (DocSimulator) in *NEJM AI*, using faculty-graded OSCE performance as the primary endpoint. At the agent-capability frontier, Tu et al. [15] introduced AMIE, a self-play–trained diagnostic-dialogue agent that matched or exceeded primary-care physicians across history taking, diagnostic reasoning, management, and patient-perceived empathy in standardized patient-actor encounters. Most recently, Kyung et al. [19] introduced PatientSim, a persona-driven simulator comprising 37 personas, validated by four clinicians. Existing LLM-VSP studies typically score model output against rubric items or encounter-level patient-actor impressions, evaluate a single backbone LLM, and rely on small or single-rater panels. The present evaluation, in contrast, compares every model utterance against an utterance-aligned response from a practicing medical educator as a 1:1 gold standard, scores every utterance under a fully-crossed assignment by seventeen OSCE-trained final-year medical-student raters, and stratifies the resulting scores along a history-taking and linguistic-form taxonomy that localizes where each model incurs Likert-scale penalties. To our knowledge, this methodological combination has not been instantiated in prior LLM-VSP research.

Multi-rater human evaluation of clinical dialogue agents. Educational simulators differ from closed-form benchmarks in that there is no single correct answer to score against; quality is multidimensional, partly subjective, and context-bound. Aggregate accuracy or single-judge LLM-as-a-judge scores, therefore, underdetermine the educational utility of a VSP. Multi-rater human evaluation has become the gold standard whenever judgments are inherently subjective: the DICES dataset [20] uses 123 raters per dialogue to model rater diversity rather than treat disagreement as noise, and HealthBench [21] engages 262 physicians from 60 countries to score 5,000 multi-turn health conversations.

Stratified, capability-based evaluation. The lineage initiated by CheckList [22] and broadened by HELM [23] argues that systems should be assessed not only on overall scores but on category-defined failure modes, and that publishing diagnostic detail is itself a contribution. Within medicine, Yao et al. [24] introduced MedQA-CS as an OSCE-style closed-form benchmark with Cohen's α agreement, and Jiang et al. [25] built MedAgentBench as an EHR-agent environment validated by two physicians. CLEVER [26] provides a methodology for blind preference-based evaluation by external clinicians. We combine 1) fully-crossed multi-rater scoring of LLM-VSP dialogue, 2) explicit history-taking and linguistic stratification of the question itself, and 3) analysis of how stratification surfaces model-specific failure modes that aggregate scores hide. The framework also transfers from static evaluation to dialogue collected on a deployed platform.

3 VSP Construction and Response Generation

A virtual standardized patient evaluated against an OSCE encounter must satisfy three requirements: it must inhabit a clinically coherent persona consistent with a specific scenario, respond to free-form student utterances in role and in real time, and do so reproducibly so that responses are comparable across systems. We construct VSPs that meet these requirements by composing OSCE scenarios

written by a clinical educator, a single declarative persona-injection prompt that instantiates each scenario, and a multi-source response corpus in which four frontier LLMs and a human medical-education expert independently produce responses to the same input utterances.

3.1 OSCE Scenarios and Input Prompts

Three OSCE scenarios were developed: hematuria (red urine), fever in a pregnant patient, and chronic cough. Each was developed from the Korean OSCE/CPX training guide for medical students [27] and refined by a board-certified internal medicine faculty member with thirteen years of experience in clinical skills assessment. The cases span a common general-medicine presentation (hematuria), a special-population case requiring sensitive history-taking (pregnancy), and a chronic presentation requiring multi-specialty differential reasoning (cough). Each scenario specifies the patient's chief complaint, demographic profile, history of present illness, past medical, drug, family, and social history, and the expected behavioral style of the SP (e.g., reluctance to disclose particular history). Scenarios are stored as structured documents that map directly to the persona-injection prompt fields used in the next subsection.

A study-specific set of seventy input prompts (23 hematuria, 23 fever, 24 cough) was prepared as the queries against which all VSPs are evaluated. These prompts are student-doctor utterances sampled from authentic OSCE-style training transcripts produced in prior cohorts at the same medical school and represent the question types that medical students typically deploy during a 10-minute OSCE encounter, namely opening greetings, chief-complaint elicitation, character and associated-symptom probing, past-medical and lifestyle inquiry, summarization, and patient education. Each prompt is independently tagged with a history-taking category and a linguistic-form label, as detailed in §4.3. All scenarios and the input-prompt set were reviewed and approved by the Institutional Review Board of Dankook University Hospital prior to data collection.

3.2 VSP Prompting via Persona Injection

Each VSP is constructed by *persona injection* [18], a system-prompt-based approach in which a single declarative prompt instructs the model to play a specific clinical role and to answer turn-by-turn from within it. The same template, comprising six fields (role declaration, demographic context, clinical state, history, conversational style, and behavioral constraints; full template in Appendix B), was instantiated three times, once per scenario, with case-specific values; no other prompt text varied between scenarios. Each instantiation is on the order of 700–900 Korean characters and is released verbatim with the dataset.

Inference parameters were held constant across the four LLMs: temperature 0.7, top-p 0.95, maximum response length 1,024 tokens, and a single response per query (no retries, no chain-of-thought scaffolding). The four LLMs evaluated were GPT-4o (OpenAI), Claude 3.5 Sonnet (Anthropic), Llama-3.1 70B Instruct (Meta, served via a HuggingFace Inference Endpoint), and HyperCLOVA X (NAVER, served via the official Korean-language API). Specific dated API releases are omitted; all four models were queried within a single calendar week to minimize drift across silent updates.

3.3 Generating the Multi-Source Response Corpus

For each of the seventy input prompts, five sources independently produced a response: the four LLMs above and one human medical-education expert (the same internal-medicine faculty member who authored the scenarios). Expert responses were composed without exposure to model outputs (write-blind condition) and at the level of detail expected from a trained standardized patient at a national OSCE workshop. The expert response set serves as a 1:1 utterance-aligned gold standard against which model outputs are directly compared.

The complete multi-source response corpus consists of 5 sources $\times$ 70 input prompts = 350 responses. Responses were stored together with their source label, randomized within case for rater presentation to mitigate position-order bias, and re-blinded such that raters did not know which response came from which source.

4 Stratified Multi-Rater Evaluation

A multi-rater human evaluation framework was applied to the 350-response corpus described above. Seventeen raters scored every response on three sentence-level metrics (Fluency, Accuracy, Contextual Relevance) and every encounter on six encounter-level metrics. The fully-crossed design produced 5,950 sentence-level rating events (17 raters $\times$ 350 responses) and 1,530 encounter-metric assessments (17 raters $\times$ 15 encounters $\times$ 6 metrics), totalling 7,480 distinct rating actions. Resulting scores were stratified along an 11-category history-taking taxonomy crossed with a 3-level linguistic-form label and analyzed using inter-rater reliability statistics and a linear mixed-effects model.

4.1 Rater Recruitment

Seventeen senior medical students were recruited from Korean medical schools via an open invitation circulated through the medical-education office (mean age 24.8 $\pm$ 1.9 years; 5 women [29.4%]; Year 6 n = 10 [58.9%], Year 5 n = 5 [29.4%], Year 4 n = 2 [11.8%] of the 6-year curriculum). All had completed at least two prior OSCE training cycles, ensuring familiarity both with the OSCE format and with the educational target of an SP-style encounter.

A 60-minute calibration session preceded the rating phase, in which the rubric (§4.2) was reviewed against three illustrative responses (one human-expert, one strong-LLM, one weak-LLM) drawn from a held-out scenario; raters discussed disagreements until anchor descriptions for each Likert level were shared. Raters then scored the full corpus asynchronously over a four-week window in randomized order and with sources de-identified.

4.2 Sentence- and Encounter-Level Metrics

Two granularities of evaluation are used (Table 1). At the sentence level (one rating per response, where a response corresponds to one VSP utterance), each rater scored Fluency, Accuracy, and Contextual Relevance on five-point Likert scales anchored to written descriptors at each level. At the encounter level (one rating per scenario $\times$ source pair), six metrics were used to capture properties that emerge only from the full conversational arc: comprehensibility, hallucination-free, coherence, consistency, engagement, and overall satisfaction. The complete rubric is reproduced in Appendix C.

Table 1: Sentence- and encounter-level evaluation metrics. The "Rater item" column shows the exact Likert agreement statement presented to raters (translated from Korean), rated on a 5-point scale (1 = strongly disagree; 5 = strongly agree).

Metric	Rater item
Sentence-level (per VSP utterance)	
Fluency	The response reads naturally, as if produced by a human.
Accuracy	The response is factually accurate based on the provided patient case description.
Contextual Relevance	In the OSCE setting, the response accurately addresses the intent of the student's question.
Encounter-level (per scenario $\times$ source pair)	
Comprehensibility	The virtual patient's responses are clear and easy to understand.
Hallucination-free	The virtual patient does not provide content not given in the case description.
Coherence	The dialogue flow is natural and coherent across turns.
Consistency	The persona is consistent across the encounter.
Engagement	The conversation feels engaging, comparable with a human SP.
Overall Satisfaction	Overall, the encounter would be acceptable as a substitute for a human SP.

4.3 Question Stratification: History-Taking Function and Linguistic Form

We adopt the stratified-evaluation paradigm, in which models are assessed on category-defined slices of input rather than aggregate scores alone, as introduced by CheckList [22] and broadened by HELM [23]. Each input prompt is annotated along two independent axes, which, when crossed, reveal stratified contrasts that aggregate scores cannot detect. The first axis is an 11-category history-taking taxonomy grounded in the OSCE history-taking structure of AMEE Guide No. 81 [1] and validated

against the systematic review of medical-student history-taking instruction by Keifenheim et al. [28] (Table 2). The second is a three-level linguistic-form label distinguishing *Affirmative* (positive-polarity questions and statements), *Negative* (negative-polarity questions), and *Open-ended* (unconstrained-response questions). The Affirmative–Negative distinction is motivated by the polarity-response divergence between Korean and English, which is the failure mode that the linguistic axis is designed to surface. Both axes were applied with a rule-based tagger; ambiguous utterances were reviewed by two faculty members and resolved by consensus. The two-axis full cross yields an 11×3 stratification grid that supports cell-level aggregation, marginal analysis, and interaction modeling.

Table 2: Eleven-category history-taking taxonomy of input prompts.

Category	# Q	Definition	Example
Greeting & ID check	3	Opening of the encounter and verification of patient identity.	*"Hello, I'm a medical student. May I have your name and age?" (hematuria)*
Chief complaint (C.C.)	3	Eliciting the principal reason for the visit.	*"Hello, what brings you in today?" (hematuria)*
History of C.C. (HPC) — Character	10	Probing the qualitative character of the present complaint.	*"What color was the red urine?" (hematuria)*
HPC — Associated symptoms	21	Eliciting symptoms accompanying the chief complaint.	*"Phlegm or runny nose along with the cough?" (cough)*
HPC — Aggravating/relieving	3	Contextual factors that worsen or improve the symptom.	*"Does the cough get worse when you lie down?" (cough)*
Past medical history	6	Previous diagnoses and prior episodes.	*"Diagnosed with diabetes before?" (fever)*
Drug history	3	Current and recent medications.	*"Taking any medications regularly?" (fever)*
Family history	2	Hereditary conditions relevant to differential.	*"Anyone in family with lung cancer?" (cough)*
Social history	9	Tobacco, alcohol, occupation, lifestyle.	*"Do you drink alcohol or smoke?" (cough)*
Clinical reasoning	2	Summarization and check-back.	*"To summarize: cold one week ago, then reddish-brown urine…" (hematuria)*
Patient education	8	Diagnosis explanation, counseling, lifestyle guidance.	*"Hematuria may be associated with smoking; I recommend quitting." (hematuria)*

4.4 Inter-Rater Reliability and Mixed-Effects Modeling

Inter-rater reliability was quantified at both sentence and encounter levels by ICC(2,k) (two-way random-effects, absolute agreement, average measures; following Koo and Li [29]) and by Krippendorff's α with ordinal weights [30]; ICC values are reported with 95% confidence intervals (F-distribution method).

Statistical comparison of model performance was carried out by a linear mixed-effects model [31] fitted to rater-level Likert scores. The model included *Source* (5 levels), *History-taking category* (11 levels), and *Linguistic form* (3 levels) as fixed effects, with *Source* $\times$ *History-taking category* and *Source* $\times$ *Linguistic form* interactions to expose stratified contrasts; *Rater* was modeled as a random intercept and *Utterance* as a random intercept nested within *Case*. Estimation was by restricted maximum likelihood (REML); estimated marginal means and contrasts between Source levels (and across the *a priori* Negative-vs-Affirmative contrast within each Source) are reported with 95% confidence intervals. Descriptive means reported in Table 3, Appendix D, and Figure E1 are accompanied by 95% nonparametric percentile bootstrap confidence intervals (2,000 resamples). All analyses were performed in R 4.5.2 and Python 3.11.

Data availability. All static-evaluation materials, namely the 350-response corpus, the 7,480 rater scores, the input-prompt set with two-axis annotations, and the persona-injection prompt template, are provided in full in the Appendices (Appendices B-C). The static evaluation set is not separately hosted as a stand-alone public dataset given its modest size and the human-expert-reference component. The platform-collected dialogue corpus is released separately as a public Hugging Face dataset under CC BY-NC-SA 4.0 with a Croissant Core and Responsible-AI metadata file at `https://huggingface.co/datasets/nwirandx/Korean-OSCE-history-taking-dialogues`. The Datasheet for Datasets (Appendix A) follows Gebru et al. [32] and documents the platform-collected dataset.

5 Experimental Results

5.1 Encounter-level performance

The human-expert reference led on all six encounter-level metrics (Table 3). Mixed-effects contrasts against the human reference (Figure 1) showed that three of the four LLMs (Claude 3.5 Sonnet, GPT-4o, HyperCLOVA X) stayed within 0.5 Likert points of the reference on Comprehensibility, Consistency, and Hallucination-freeness. In four of the 24 source $\times$ metric cells the contrast was not statistically significant: Hallucination-freeness in Claude 3.5 Sonnet and GPT-4o, Consistency in GPT-4o, and Engagement in HyperCLOVA X (-0.02 [-0.31, 0.27]). Llama-3.1 70B fell significantly below the reference on all six metrics and trailed every other source, with the largest gap on Coherence (-1.86 [-2.14, -1.59]) and Overall Satisfaction showing an additional 1.5–2 point spread below the reference.

Table 3: Encounter-level mean Likert score [95% bootstrap CI] for each source, pooled across the three OSCE cases. Each cell pools 51 ratings (17 raters $\times$ 3 cases).

Source	Compre-hensibility	Hallucination-free	Coherence	Consistency	Engagement	Overall satisfaction
Human (reference)	**4.84 [4.74–4.94]**	**4.49 [4.29–4.67]**	**4.80 [4.69–4.92]**	**4.75 [4.57–4.88]**	**4.18 [3.92–4.43]**	**4.61 [4.41–4.78]**
Claude 3.5 Sonnet	4.45 [4.27–4.63]	4.35 [4.12–4.55]	4.00 [3.75–4.25]	4.39 [4.16–4.61]	3.82 [3.55–4.10]	3.82 [3.53–4.12]
GPT-4o	4.59 [4.39–4.76]	4.41 [4.22–4.61]	4.08 [3.84–4.29]	4.53 [4.35–4.71]	3.75 [3.49–3.98]	3.82 [3.57–4.08]
Llama-3.1 70B	3.71 [3.43–3.98]	3.94 [3.69–4.18]	2.94 [2.67–3.24]	3.82 [3.53–4.14]	2.94 [2.63–3.25]	2.71 [2.35–3.06]
HyperCLOVA X	4.57 [4.41–4.71]	4.02 [3.78–4.25]	4.24 [4.04–4.43]	4.39 [4.24–4.55]	4.16 [3.92–4.37]	3.88 [3.65–4.10]

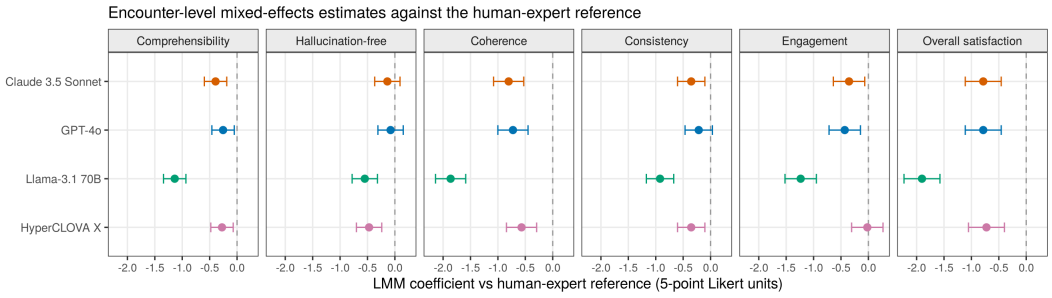


Figure 1: Linear mixed-effects model estimates (mean difference vs. human-expert reference, in 5-Likert units) with 95% confidence intervals for each LLM $\times$ encounter-level metric. Negative values indicate lower than the human reference; the reference (Human) sits at 0 by construction.

5.2 Sentence-level performance

Sentence-level performance was substantially higher and more compressed across sources than encounter-level (Figure 2). Pooled Accuracy was within 0.09 Likert points of the human reference across all four LLMs, indicating that within-utterance factual alignment is the most achievable property. Fluency separated the sources most clearly: human reference 4.91, HyperCLOVA X 4.68, GPT-4o 4.39, Claude 3.5 Sonnet 4.31, Llama-3.1 70B 3.69 of 5. The Fluency ordering anticipates the encounter-level Coherence ordering (Figure 1), suggesting that local linguistic naturalness propagates into global encounter quality.

5.3 Stratified analysis by question typology

Question-typology stratification revealed model- and category-specific failure modes that aggregate scores obscured. History-taking category $\times$ Source interactions (Figure 3) identified pronounced LLM weaknesses on Clinical reasoning and Patient education, categories that require synthesis across the prior dialogue rather than direct symptom recall. They also revealed that Llama-3.1 70B's deficit was largest on Greeting & ID check and Clinical reasoning, exceeding 1 Likert point below the human reference on Fluency. Linguistic-form $\times$ Source interactions surfaced a Korean negative-form penalty on Fluency. Relative to the human reference, the additional drop from Affirmative-form questions

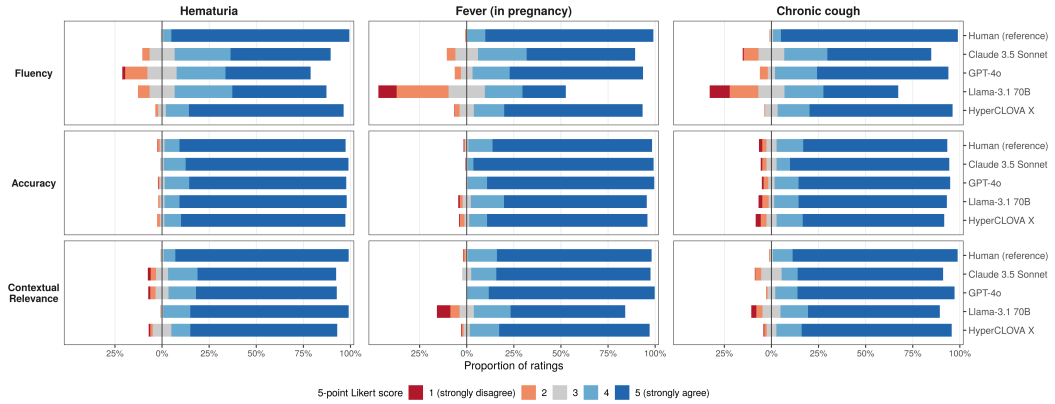


Figure 2: Diverging stacked-bar distribution of Likert ratings for the three sentence-level metrics for each of the five sources, shown for each of the three OSCE cases. The vertical zero line splits ratings into negative (1–2 plus half of 3) on the left and positive (4–5 plus half of 3) on the right.

to negative-polarity yes/no questions was 0.149 [-0.286, -0.011] for Llama-3.1 70B, 0.124 [-0.261, 0.014] for GPT-4o, 0.093 [-0.231, 0.044] for HyperCLOVA X, and 0.033 [-0.170, 0.105] for Claude 3.5 Sonnet; the human reference itself showed no such directional effect (+0.043 [-0.067, 0.153]). Among the four models, only the Llama-3.1 70B interval excludes zero; the remaining three are directional but not statistically significant. The ordering does not track Korean-language ability: the penalty is smallest for Claude 3.5 Sonnet and larger for HyperCLOVA X, the only Korean-developed model. The effect is one of question form, not of proficiency. Per-form means by source are visualized in Appendix E (Figure E1).

Sixteen of the 350 utterances fell below 3.0 on at least one sentence-level metric, and fourteen of these were Fluency failures. In seven utterances, all in Clinical reasoning or Patient education, the virtual patient restated the student's explanation in clinical register instead of answering as a patient. In three, all negative-polarity questions, it answered a question that had not been asked. No human-reference utterance fell below 3.0 on any metric. These patterns thus distinguish the model-generated responses from the human reference.

5.4 Inter-rater reliability

Encounter-level pooled ICC(2,k) ranged from 0.79 (Hallucination-free) to 0.94 (Coherence), with a fully-pooled ICC(2,k) of 0.91 [0.88–0.94]. Sentence-level pooled ICC(2,k) per source ranged from 0.80 to 0.92. Krippendorff's α with ordinal weights was lower than ICC at both granularities (0.08–0.41), reflecting α's stricter accounting for chance agreement on ordinal scales; the two measures produced the same source-level ordering. Full per-metric and per-source values are reported in Appendix E (Table E1).

6 VSP-OSCE Training Platform

6.1 Platform architecture and scenario pipeline

The **VSP-OSCE training platform** is a web-based application that delivers persona-injected VSPs as a continuously updating educational resource and dataset pipeline. The platform invokes GPT-4.1 for clinical-scenario script generation (offline, prior to deployment) and GPT-4o for the two real-time roles, namely automated evaluation and patient role-play (chosen for its lower latency during live student sessions); all calls use seed = 42. Scenario content is derived from a structured database of 80 core symptoms commonly taught in medical curricula, linked to expert-validated disease lists and three patient profiles per symptom (240 profiles in total) covering demographics, vital signs, and diagnostic context, organized in hierarchical JSON, with symptoms at the top, diseases nested under each symptom, and patient profiles nested under each disease. For each disease, a symptom-focused history-taking checklist is generated by GPT-4.1 (temperature 0.2) under a one-shot prompt and refined through human-in-the-loop expert review; a free-form patient narrative

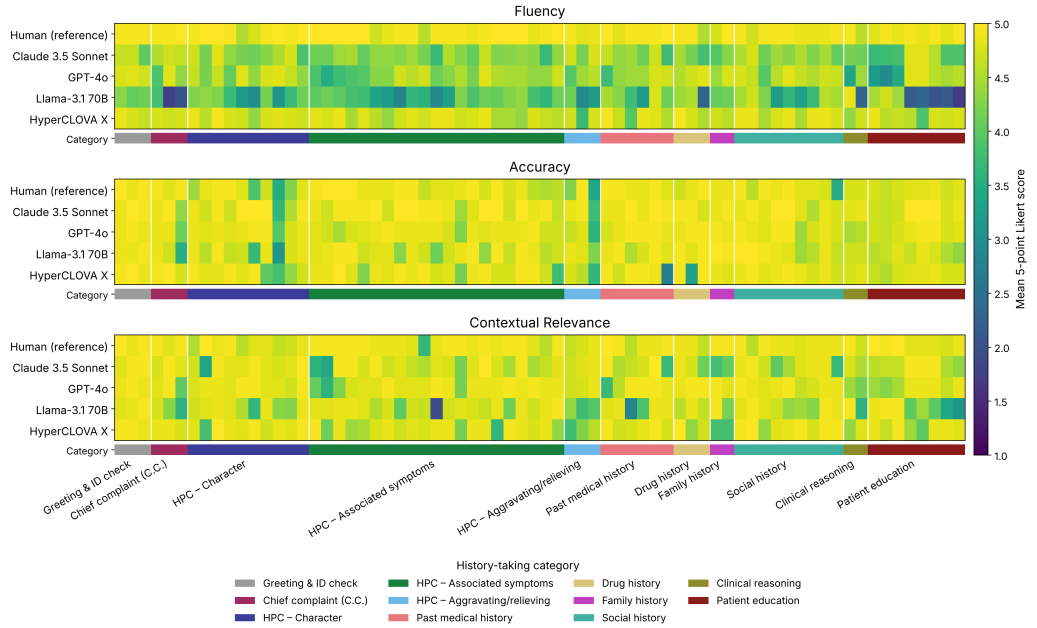


Figure 3: Sentence-level mean Likert score per question × source, organized by sentence-level metric (Fluency, Accuracy, Contextual Relevance). Within each metric panel, rows are the five sources (Human reference + four LLMs) and columns are the 70 input prompts sorted by their history-taking category; cell shading shows mean Likert score (1–5). The horizontal color bar (*Category*) below each metric panel maps each prompt to its history-taking category (legend at bottom).

($\geq$ 1,500 characters) is then generated and verified by a downstream diagnostic model that retains the scenario only when its predicted disease matches the intended diagnosis. At session start, the verified scenario is loaded into the persona-injection prompt template (Appendix B) and instantiated by GPT-4o (temperature 0.5, max 35 tokens), with the 35-token cap constraining responses to one or two sentences. An automated GPT-4o checklist-based score (temperature 0), computed at session close by matching the student's questions against the scenario-specific history-taking checklist, is shown to the participant for self-directed practice. The platform code is publicly available at `https://github.com/JINAILAB/OSCE-SimBench`.

6.2 Real-time interaction and released dataset

Once a participant logs in and selects a scenario, the platform hands control to the participant, who types a free-form history-taking question via a real-time text interface. The platform routes each question to GPT-4o (VSP role-play), which returns a layperson-language response constrained by behavioral rules: avoiding unsolicited information, medical jargon, and expressions of certainty when asked about diagnoses. All turns are appended to a session log in chronological order, and each session is anonymized and stored as a structured record (`session_id`, `participant_id` R001–R017, `clinical_scenario`, `turn_no`, `student_question`, `vsp_response`).

To date, the 17 medical-student raters who scored the static evaluation (§4.1) used the platform during a pilot phase, generating 49 sessions across 27 distinct clinical scenarios between August 2025 and May 2026, totaling approximately 1,763 student–VSP turns. The platform-collected dialogue corpus is released as a public Hugging Face dataset (`https://huggingface.co/datasets/nwirandx/Korean-OSCE-history-taking-dialogues`) under CC BY-NC-SA 4.0 with a Croissant metadata file containing both Core fields (auto-generated by Hugging Face) and Responsible-AI fields specifying limitations, biases, intended use cases, social impact, synthetic-data flags, and provenance. Applying the history-taking taxonomy of §4.3 to the platform dialogue confirms that the eleven categories generalize beyond the static 70-prompt evaluation set: all eleven are represented among the 1,614 platform utterances the scheme covers, with the per-category distribution reported in Appendix F. The platform thus closes the loop between static evaluation methodology and continuously growing

evaluation resource, supporting medical education (immediate practice with feedback) and providing a privacy-compliant dataset pipeline for downstream LLM-VSP research.

7 Discussion

Limitations. This study has several limitations. First, the human-expert reference responses were composed by a single OSCE-faculty author. This approach kept the reference internally consistent across the three cases, but limits its representativeness of OSCE educators and trained standardized patients more broadly. Where a model did not differ significantly from the reference, that result is therefore specific to this expert. Raters were also not asked to judge whether a response was human-authored, so whether they could have distinguished virtual from human responses is not recoverable from these data. Second, the burden of a fully-crossed human rating constrained the rater pool to 17 senior medical students and the LLM panel to four models. Several recently released open-source frontier models that postdated the rating window were therefore excluded from the comparison. Third, the simulator is text-only and cannot reproduce prosody, facial expression, or gesture, so encounter-level properties such as Engagement and Comprehensibility may behave differently in spoken settings than the values reported here. Fourth, the three scenarios sample only a portion of the OSCE assessment. Korean OSCE/CPX case banks comprise more than fifty case classes, and specialized settings such as psychiatry and pediatrics are not represented in the present set.

Future directions. Several extensions follow from the present work. First, broadening the human-reference pool by including several OSCE educators across specialties, trained standardized patients, and more recent commercial and open-source LLMs would strengthen the utterance-level gold standard and improve the precision of any comparison drawn against it. Second, the 7,480 fully-crossed multi-rater scores released here can serve as a validation set for developing LLM-as-a-judge evaluation pipelines, which could substantially reduce rater burden if validated against human ratings. Third, extending the platform beyond text to speech, video, or VR-based simulators would yield richer student–VSP encounters and engagement signals closer to those of a real OSCE.

In summary, this study introduces a multi-rater human-evaluation framework, a persona-injection prompt set, and a deployed VSP platform that together cover scenario authoring, evaluation, and continuous data generation for LLM-based clinical dialogue. The framework supports both medical-education practice and downstream LLM evaluation, and is released to facilitate further research on virtual standardized patients in OSCE-based training settings. Establishing the educational effect of VSP-based OSCE practice will require a controlled comparative study of learning outcomes.

Acknowledgments and Disclosure of Funding

This work was supported by the Bio&Medical Technology Development Program of the National Research Foundation (NRF), funded by the Korean government (MSIT) (RS-2023-00220408), and by the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (RS-2025-02307233). Additional support was provided by the Center for Artificial Intelligence in Healthcare at Seoul National University Bundang Hospital (computing infrastructure and technical advice), by the NAVER Cloud team (API credits and access to HyperCLOVA X), and by an unrestricted donation from Dr. Sang-Gu Lee, director of Seoul Top Internal Medicine Clinic. None of these funders, donors, or providers had any role in the study design, the evaluation protocol, the analysis, or the interpretation of the results. The authors declare no competing interests.

References

[1] K. Z. Khan, S. Ramachandran, K. Gaunt, and P. Pushkar. The Objective Structured Clinical Examination (OSCE): AMEE Guide No. 81. Part I: an historical and theoretical perspective. *Medical Teacher*, 35:e1437–e1446, 2013. doi: 10.3109/0142159X.2013.818634. `https://doi.org/10.3109/0142159X.2013.818634`

[2] K. S. Kim. Introduction and administration of the clinical skill test of the medical licensing examination, Republic of Korea (2009). *Journal of Educational Evaluation for Health Professions*, 7:4, 2010. doi: 10.3352/jeehp.2010.7.4. `https://doi.org/10.3352/jeehp.2010.7.4`

[3] C.-W. Lin, T.-C. Tsai, C.-K. Sun, D.-F. Chen, and K.-M. Liu. Power of the policy: how the announcement of high-stakes clinical examination altered OSCE implementation at institutional level. *BMC Medical Education*, 13:8, 2013. doi: 10.1186/1472-6920-13-8. https://doi.org/10.1186/1472-6920-13-8

[4] S. Müller, I. Koch, U. Settmacher, and U. Dahmen. How the introduction of OSCEs has affected the time students spend studying: results of a nationwide study. *BMC Medical Education*, 19:146, 2019. doi: 10.1186/s12909-019-1570-6. https://doi.org/10.1186/s12909-019-1570-6

[5] R. Yudkowsky and K. Szauter. Farewell to the Step 2 Clinical Skills exam: new opportunities, obligations, and next steps. *Academic Medicine*, 96:1250–1253, 2021. doi: 10.1097/ACM.0000000000004209. https://doi.org/10.1097/ACM.0000000000004209

[6] L. Vogel. Controversial medical licensing exam cancelled. *CMAJ*, 192:E1417–E1418, 2020. doi: 10.1503/cmaj.1095904. https://doi.org/10.1503/cmaj.1095904

[7] L. Vogel. Medical Council of Canada abandons controversial exam. *CMAJ*, 193:E1014–E1015, 2021. doi: 10.1503/cmaj.1095947. https://doi.org/10.1503/cmaj.1095947

[8] B. S. Malau-Aduli, K. Jones, S. Saad, and C. Richmond. Has the OSCE met its final demise? Rebalancing clinical assessment approaches in the peri-pandemic world. *Frontiers in Medicine*, 9:825502, 2022. doi: 10.3389/fmed.2022.825502. https://doi.org/10.3389/fmed.2022.825502

[9] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, et al. Large language models encode clinical knowledge. *Nature*, 620:172–180, 2023. doi: 10.1038/s41586-023-06291-2. https://doi.org/10.1038/s41586-023-06291-2

[10] D. Brin, V. Sorin, A. Vaid, A. Soroush, B. S. Glicksberg, A. W. Charney, G. Nadkarni, and E. Klang. Comparing ChatGPT and GPT-4 performance in USMLE soft skill assessments. *Scientific Reports*, 13:16492, 2023. doi: 10.1038/s41598-023-43436-9. https://doi.org/10.1038/s41598-023-43436-9

[11] F. Holderried, C. Stegemann-Philipps, L. Herschbach, J.-A. Moldt, A. Nevins, J. Griewatz, M. Holderried, A. Herrmann-Werner, T. Festl-Wietek, and M. Mahling. A generative pretrained transformer (GPT)-powered chatbot as a simulated patient to practice history taking: prospective, mixed methods study. *JMIR Medical Education*, 10:e53961, 2024. doi: 10.2196/53961. https://doi.org/10.2196/53961

[12] A. Elhilali, A. S.-H. Ngo, D. Reichenpfader, and K. Denecke. Large language model-based patient simulation to foster communication skills in health care professionals: user-centered development and usability study. *JMIR Medical Education*, 11:e81271, 2025. doi: 10.2196/81271. https://doi.org/10.2196/81271

[13] E. Brügge, S. Ricchizzi, M. Arenbeck, M. N. Keller, L. Schur, W. Stummer, M. Holling, M. H. Lu, and D. Darici. Large language models improve clinical decision making of medical students through patient simulation and structured feedback: a randomized controlled trial. *BMC Medical Education*, 24:1399, 2024. doi: 10.1186/s12909-024-06399-7. https://doi.org/10.1186/s12909-024-06399-7

[14] E. Lavigne, A. Lopez, J. Frandon, G. Blaizot, L. Gabellier, S. Adham, J. U. Bedoya, J. Charriot, T. Thieblemont, M. Badreddine, E. Guenou, F. Carbonnel, C. Duflos, D. Morin, I. Laffont, Y.-M. Pers, and K. Yauy. AI-standardized clinical examination training on OSCE performance. *NEJM AI*, 2(8):AIoa2500066, 2025. doi: 10.1056/AIoa2500066. https://doi.org/10.1056/AIoa2500066

[15] T. Tu, M. Schaekermann, A. Palepu, K. Saab, J. Freyberg, R. Tanno, A. Wang, B. Li, M. Amin, Y. Cheng, et al. Towards conversational diagnostic artificial intelligence. *Nature*, 642:442–450, 2025. doi: 10.1038/s41586-025-08866-7. https://doi.org/10.1038/s41586-025-08866-7

[16] J. Zeng, W. Qi, S. Shen, X. Liu, S. Li, B. Wang, J. Yu, J. Lin, Y. Liu, J. Lu, X. Wei, X. Tu, J. Wu, X. Zhang, and Y. Lin. Embracing the future of medical education with large language model-based virtual patients: scoping review. *Journal of Medical Internet Research*, 27:e79091, 2025. doi: 10.2196/79091. https://doi.org/10.2196/79091

[17] D. Li and S. Lebai Lutfi. Large language model-based virtual patient systems for history-taking in medical education: comprehensive systematic review. *JMIR Medical Informatics*, 14:e79039, 2026. doi: 10.2196/79039. https://doi.org/10.2196/79039

[18] Y.-M. Tseng, Y.-C. Huang, T.-Y. Hsiao, W.-L. Chen, C.-W. Huang, Y. Meng, and Y.-N. Chen. Two tales of persona in LLMs: a survey of role-playing and personalization. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 16612–16631, 2024. doi: 10.18653/v1/2024.findings-emnlp.969. https://doi.org/10.18653/v1/2024.findings-emnlp.969

[19] D. Kyung, H. Kim, J. Lee, E. Kang, and E. Choi. PatientSim: A persona-driven simulator for realistic doctor-patient interactions. arXiv:2505.17818, 2025. https://arxiv.org/abs/2505.17818

[20] L. Aroyo, A. S. Taylor, M. Diaz, C. M. Homan, A. Parrish, G. Serapio-Garcia, V. Prabhakaran, and D. Wang. DICES dataset: diversity in conversational AI evaluation for safety. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023) Datasets and Benchmarks Track*, 2023. https://arxiv.org/abs/2306.11247

[21] R. K. Arora, J. Wei, R. Soskin Hicks, P. Bowman, et al. HealthBench: Evaluating large language models towards improved human health. arXiv:2505.08775, 2025. https://arxiv.org/abs/2505.08775

[22] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh. Beyond accuracy: behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912, 2020. doi: 10.18653/v1/2020.acl-main.442. https://doi.org/10.18653/v1/2020.

acl-main.442

[23] R. Bommasani, P. Liang, T. Lee, et al. Holistic evaluation of language models. *Annals of the New York Academy of Sciences*, 1525:140–146, 2023. doi: 10.1111/nyas.15007. https://doi.org/10.1111/nyas.15007

[24] Z. Yao, Z. Zhang, C. Tang, X. Bian, Y. Zhao, Z. Yang, J. Wang, H. Zhou, W. S. Jang, F. Ouyang, and H. Yu. MedQA-CS: Objective Structured Clinical Examination (OSCE)-style benchmark for evaluating LLM clinical skills. arXiv:2410.01553, 2024. https://arxiv.org/abs/2410.01553

[25] Y. Jiang, K. C. Black, G. Geng, D. Park, A. Y. Ng, and J. H. Chen. MedAgentBench: A realistic virtual EHR environment to benchmark medical LLM agents. *NEJM AI*, 2025. doi: 10.1056/AIdbp2500144. https://doi.org/10.1056/AIdbp2500144

[26] V. Kocaman, M. A. Kaya, A. M. Feier, and D. Talby. Clinical large language model evaluation by expert review (CLEVER): framework development and validation. *JMIR AI*, 4:e72153, 2025. doi: 10.2196/72153. https://doi.org/10.2196/72153

[27] H. Roh, K. M. Lee, E. Eo, Y. S. Hong, H. Lee, B. W. Jang, and B. D. Rhee. Development of guide to clinical performance and basic clinical skills for medical students. *Korean Journal of Medical Education*, 27(4):309–319, 2015. doi: 10.3946/kjme.2015.27.4.309. https://doi.org/10.3946/kjme.2015.27.4.309

[28] K. E. Keifenheim, M. Teufel, J. Ip, N. Speiser, E. J. Leehr, S. Zipfel, and A. Herrmann-Werner. Teaching history taking to medical students: a systematic review. *BMC Medical Education*, 15(1):159, 2015. doi: 10.1186/s12909-015-0443-x. https://doi.org/10.1186/s12909-015-0443-x

[29] T. K. Koo and M. Y. Li. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15(2):155–163, 2016. doi: 10.1016/j.jcm.2016.02.012. https://doi.org/10.1016/j.jcm.2016.02.012

[30] K. Krippendorff. *Content Analysis: An Introduction to Its Methodology*. Sage Publications, 2nd edition, 2004.

[31] D. Bates, M. Mächler, B. Bolker, and S. Walker. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1):1–48, 2015. doi: 10.18637/jss.v067.i01. https://doi.org/10.18637/jss.v067.i01

[32] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. (Originally as arXiv:1803.09010, 2018). https://doi.org/10.1145/3458723

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 list three explicit contributions (multi-rater human evaluation, question-stratified evaluation, deployed VSP platform); each is directly demonstrated in §5 (results), §6 (platform), and the Appendices.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §7 explicitly enumerates three limitations: single-author human reference, constrained scale (n = 17 raters), and text-only modality.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper presents an empirical evaluation and dataset release; no theoretical results are claimed.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §3–§4 specify all evaluation protocol details (LLM model identities, persona-injection prompt template, rater calibration procedure, statistical model). The full persona-injection prompt is reproduced in Appendix B; the response corpus and rater scores are provided in Appendices C and D.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: All static-evaluation materials (350-response corpus, 7,480 rater scores, input-prompt set with annotations, persona-injection prompt template) are provided in full in Appendices B–E. The platform-collected dialogue corpus is released as a public Hugging Face dataset under CC BY-NC-SA 4.0 with Croissant Core and Responsible-AI metadata; the platform code is publicly available at `https://github.com/JINAILAB/OSCE-SimBench`.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: §3 specifies inference parameters (temperature, top-p, max tokens, single response per query); §4.4 specifies the statistical model (LMM with random effects for Rater and Utterance|Case, REML estimation); §5 reports 95% bootstrap CIs and 95% CIs for ICC.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: All effect estimates are reported with 95% confidence intervals; bootstrap CIs are used for descriptive means; F-distribution method for ICC; LMM contrasts use Wald-style 95% CIs.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [N/A]

   Justification: The paper evaluates externally hosted LLM endpoints; no model training or fine-tuning was performed. Inference incurred negligible compute cost (single response per prompt across 350 prompts $\times$ 4 LLMs).

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research conforms to the NeurIPS Code of Ethics: human raters were recruited voluntarily under written informed consent and IRB approval; compensation was at the institution's standard research-participation rate.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §1 and §6 describe positive societal impact (cost-effective OSCE training resource accessible across institutions). Negative-impact considerations (e.g., over-reliance on automated VSPs, language-coverage limitations) are addressed in §7 limitations and the Datasheet (Appendix A).

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The released artefacts are evaluation data (rater scores, response transcripts) and dialogue logs; no high-risk pretrained models or generative checkpoints are released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Every cited LLM, dataset, and external software package is named in the manuscript with its source and licensing terms summarized in the Datasheet (Appendix A); the released platform-collected dataset is licensed under CC BY-NC-SA 4.0.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The platform-collected dialogue corpus is documented via a Datasheet for Datasets (Appendix A) and a Croissant metadata file with Core and Responsible-AI fields; intended use, limitations, and biases are described.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [Yes]

    Justification: Seventeen senior medical-student raters were recruited via an open invitation circulated through the medical-education office; they completed a 60-minute calibration session before scoring; participants received compensation at the institution's standard student research-participation rate. Rating instructions (the Likert agreement statements presented to raters) are reproduced in Table 1 and Appendix C.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [Yes]

Justification: All scenarios, the input-prompt set, and the rating protocol were reviewed and approved by the Institutional Review Board of Dankook University Hospital prior to data collection. Participation was voluntary, written informed consent was obtained, and the platform-collected dataset is released under research-participation and personal-information consent.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs (GPT-4o, Claude 3.5 Sonnet, Llama-3.1 70B, HyperCLOVA X) constitute the experimental subjects of this evaluation; their roles and configurations are fully described in §3.3 (response generation) and §6.1 (platform deployment).

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

A Datasheet for Datasets

This Datasheet documents the platform-collected dialogue corpus released on Hugging Face. It follows the seven-section template of Gebru et al. (2018). The static evaluation materials referenced in the main paper (the 350-response corpus, 7,480 rater scores, and persona-injection prompt template) are provided in Appendices B–F and are not part of the publicly hosted dataset described here.

A.1 Motivation

For what purpose was the dataset created? To support research on LLM-based VSPs in Korean OSCE-style medical education by capturing real-time student–VSP dialogue collected on the deployed VSP-OSCE training platform.

Who created the dataset? The dataset was created by the authors of this work. The Likert ratings were contributed by 17 senior medical-student raters recruited with informed consent; the raters are not authors of this work.

Who funded the creation of the dataset? The Bio&Medical Technology Development Program of the National Research Foundation (NRF) funded by the Korean government (MSIT) (RS-2023-00220408).

Any other comments? Created under IRB-approved informed consent at Dankook University Hospital.

A.2 Composition

What do the instances that comprise the dataset represent? Single turn-pairs from real-time student–VSP encounters: one student question (human-typed in Korean) and one VSP response (GPT-4o under persona injection).

How many instances are there in total? 49 sessions across 27 scenarios,
1,763 turn-pairs from 17 medical-student participants.

Does the dataset contain all possible instances or is it a sample? All turn-pairs generated by the cohort during 2025-08 to 2026-05; not a subsample.

What data does each instance consist of? Fields: session_id, participant_id (R001–R017), clinical_scenario, turn_no, student_question, vsp_response. Per-participant summary statistics provided separately.

Is there a label or target associated with each instance? None. Raw dialogue.

Is any information missing from individual instances? No structural fields missing; a few sessions terminated early but all collected turns are retained.

Are relationships between individual instances made explicit? Yes — via session_id (groups turns) and participant_id (links sessions).

Are there recommended data splits? None.

Are there any errors, sources of noise, or redundancies? Occasional typing errors and rare role-inconsistent VSP responses; no redundancies.

Is the dataset self-contained, or does it link to external resources? Self-contained; persona-injection prompt provided as accompanying file (also in Appendix B).

Does the dataset contain confidential data? No. Identifiers replaced with R001–R017; mapping destroyed.

Does the dataset contain offensive, insulting, or harmful content? No.

Does the dataset relate to people? Yes — 17 medical-student participants at Korean medical schools (clinical-clerkship level), all anonymized.

Does the dataset identify any subpopulations? Year-of-training and aggregate metrics released; no fine-grained demographics or PHI.

Is it possible to identify individuals from the dataset? No.

Does the dataset contain sensitive data? No clinical PHI; scenarios are synthetic OSCE cases.

A.3 Collection process

How was the data acquired? Via the VSP-OSCE platform: GPT-4.1 (T=0.2) for scenario generation, GPT-4o (T=0) for automated evaluation, GPT-4o (T=0.5, max 35 tokens) for patient role-play under persona injection — all with seed=42. Dataset records the role-play output.

What mechanisms or procedures were used to collect the data? Real-time text-based interaction logged automatically; identifiers anonymized post-session.

If the dataset is a sample from a larger population, what was the sampling strategy? Convenience sample of 17 medical-student volunteers.

Who collected the data, and how were they compensated? Research team via the platform; participants paid at standard student research-participation rate.

Over what timeframe was the data collected? 2025-08-01 to 2026-05-03.

Were any ethical review processes conducted? Yes — IRB approval at Dankook University Hospital.

Was the data collected directly from individuals? Yes.

Were individuals notified about the collection? Yes — written information including purpose, anonymization, and right to withdraw.

Did the individuals consent to the collection and use of their data? Yes — research-participation and personal-information consent in writing.

Was a mechanism provided to revoke consent? Yes — withdrawn participants are removed from subsequent dataset releases.

Has an analysis of the potential impact of the dataset and its use on data subjects been conducted? IRB review served as the ethics analysis; anonymization is the principal mitigation.

Does the dataset contain synthetic data? Yes — VSP-side answers (
1,763 turns) are GPT-4o-generated under persona injection. Student questions are human-typed.

Was the dataset derived from another dataset? No external dataset. 27 scenarios authored by the research team from an 80-symptom $\times$ 3-profile internal database; verified by a diagnostic-judgment model + expert review.

A.4 Preprocessing, cleaning, and labeling

Was any preprocessing or cleaning of the data performed? Anonymization: identifiers replaced with R001–R017; in-text real names substituted; original mapping destroyed.

Was the raw data saved in addition to the preprocessed data? No.

Is the software used to preprocess the data available? Yes — released with the dataset.

Were any annotations or labels added beyond the raw dialogue? No additional labels beyond the structured-record fields.

A.5 Uses

Has the dataset been used for any tasks already? Yes — referenced in §6 of the present paper.

Is there a repository linking to all uses of the dataset? No central repository of downstream uses is maintained. Known uses will be listed on the dataset card at `https://huggingface.co/datasets/nwirandx/Korean-OSCE-history-taking-dialogues`.

What other tasks could the dataset be used for? LLM-VSP development; OSCE training research; clinical-dialogue benchmarking; post-training and preference optimization of domain-specialized LLMs.

Is there anything about the composition or collection that might affect future uses? Korean-language; senior medical students (not clinicians); text-only; 27 internal-medicine scenarios. Validity established for LLM-VSP research, OSCE training, and multi-rater methodology; NOT for clinical decision-making, diagnosis, non-Korean contexts, or standalone training.

Are there tasks for which the dataset should not be used? Clinical decision-making; patient diagnosis; standalone medical training without supervision.

Other comments on social impact? Positive: the dataset enables cost-effective, reproducible OSCE practice for medical students and supports research on LLM-based clinical dialogue agents. Negative: overreliance on LLM patient simulation could displace human standardized-patient programs, and high LLM scores could be misread as clinical competence rather than role-play fidelity. Mitigations: explicit unsuitable-use-case statements, IRB-approved consent, and full participant anonymization.

A.6 Distribution

Will the dataset be distributed to third parties outside the research team? Yes — public Hugging Face dataset under IRB-approved consent.

How will the dataset be distributed? Hugging Face at `https://huggingface.co/datasets/nwirandx/Korean-OSCE-history-taking-dialogues` with a Croissant Core + RAI metadata file.

When will the dataset be distributed? On paper acceptance; reviewers may access via the URL above.

Will the dataset be released under a copyright or other IP license? CC BY-NC-SA 4.0 (attribution + non-commercial + share-alike).

Are there other restrictions on the use of the dataset? No re-identification; no use for clinical decision-making, diagnosis, or standalone medical training.

Are there export controls or other regulatory restrictions? None known.

A.7 Maintenance

Who is supporting / hosting / maintaining the dataset? The authors of this work; hosted on Hugging Face.

How can the owner / curator / manager be contacted? `yongjin.neph@gmail.com`.

Is there an erratum? None at submission; tracked via GitHub Issues + dataset versioning.

Will the dataset be updated? Yes — new sessions appended as cohorts use the platform.

Are there limits on retention of data associated with individuals? No technical limit; withdrawn participants are removed.

Will older versions of the dataset be supported? Yes — version-tagged on Hugging Face.

Is there a mechanism for others to extend or build on the dataset? Yes — PRs on GitHub or extension issues on Hugging Face; MIT-licensed code allows third-party deployment subject to local IRB approval.

Other comments? Part of a cyclic pipeline (construction → evaluation → training → data generation) for sustainable virtual OSCE education and LLM post-training.

B Persona-Injection Prompt

The persona-injection prompt used for all three OSCE cases follows a single shared template (English translation shown below; the Korean originals are released alongside the dataset). The three case-specific instantiations differ only in the chief-complaint phrase, patient identifier, and scenario body; the surrounding instructions are otherwise identical, with one optional case-specific behavioral directive used only for the pregnancy case. We illustrate the template with one fully-instantiated worked example (hematuria); the remaining two case instantiations follow the same structure.

B.1 Unified template (English translation)

> The following information is the prior medical record of a patient presenting with [CHIEF_COMPLAINT] at the clinical-skills licensure examination. Based on this information, conduct a conversation between the [PATIENT_DESCRIPTOR] and a student-doctor.
>
> [SCENARIO_BODY — case-specific clinical narrative covering patient demographics, chief-complaint trajectory, associated symptoms, past medical / family / social history, and patient concerns; an explicit list of negative findings is included to keep VSP responses scenario-grounded.]
>
> You are now the simulated patient [PATIENT_NAME]. Please answer the questions I ask in fewer than 30 words. All answers must be complete sentences. Do not answer content that has not been asked. [OPTIONAL_CASE_SPECIFIC_DIRECTIVE]

B.2 Per-case variable assignments

Table B1: Per-case variable assignments for the unified persona-injection template (English).

Variable	Hematuria	Fever (pregnancy)	Chronic cough
CHIEF_COMPLAINT	red urine	fever (pregnant patient)	chronic cough
PATIENT_DESCRIPTOR	patient with red urine	pregnant patient	patient with chronic cough
PATIENT_NAME	Mr. Jung Jin-su (28-year-old male)	Ms. Lee Su-jin (28-year-old female, 20-week gestation)	Ms. Lee Yun-jung (55-year-old female)
OPTIONAL_DIRECTIVE	—	When the student-doctor explains that antibiotic treatment is needed, the patient should explicitly ask whether antibiotics may be harmful to the unborn child.	—

B.3 Worked example — hematuria case (English translation)

The following information is the prior medical record of a patient presenting with red urine at the clinical-skills licensure examination. Based on this information, conduct a conversation between the patient with red urine and a student-doctor.

Mr. Jung Jin-su is a 28-year-old male. He came to the hospital because of red urine that started three days ago. He had cold symptoms beginning a week ago, and on the day before the red urine appeared his temperature rose to 39 °C with associated muscle aches. The muscle aches and fever were most severe on the evening four days ago; the next morning (three days ago) he began passing red urine. The color was dark reddish-brown — initially deep reddish-brown, gradually lightening to dark amber today. He has not previously been told he had hematuria, proteinuria, or kidney disease. He takes no regular medications, herbal supplements, or health products. He has been told his blood pressure is somewhat elevated but did not consider it serious and has never visited a hospital; he does not remember the exact value. This is the first time his urine has appeared abnormal. He had no abdominal pain, flank pain, or pain at the urethral tip while passing red urine. He thinks there might have been a slight foam in the urine, but he is not sure.

Otherwise, he has not felt weak, his weight has not changed, and he denies headache, dizziness, fever, chills, muscle pain, night sweats, cough, sputum, runny nose, chest pain, dyspnoea, palpitations, exertional dyspnoea, abdominal pain, nausea, vomiting, diarrhoea, constipation, haematemesis, melaena, or haematochezia. His appetite is normal.

Mr. Jung has considered himself healthy and has rarely visited hospitals. At his early-twenties military medical examination and last year's workplace check-up, no abnormality other than slightly elevated blood pressure was noted. He has smoked about one pack a day since his military service. He drinks beer one or two glasses only at company gatherings. He has worked as an office worker for two years and undergoes annual workplace health check-ups. He used to exercise at the gym for one hour each day after work but has not gone since catching the cold a week ago. He prefers meat, eats a slightly salty diet, and does not drink much water. He works 9 AM to 6 PM and lives nearby with his family. He has a girlfriend with whom he has sex about once a week, always using a condom. He graduated from a four-year university and served in the military during his sophomore year. His father has hypertension and is on antihypertensive medication; his mother and one older brother have no known health problems.

Mr. Jung is worried about the red urine. After searching online, he found that it is called hematuria and may indicate stones or cancer; he is very concerned that he may have a serious illness. He wonders whether tests will be done today and whether he will see the results, and is also curious whether hospitalisation or detailed examinations will be required.

You are now the simulated patient Mr. Jung Jin-su. Please answer the questions I ask in fewer than 30 words. All answers must be complete sentences. Do not answer content that has not been asked.

C Translated Generated Dialogues

This appendix provides English translations of the full 350-response evaluation corpus referenced in the main paper: 70 input prompts $\times$ 5 sources (four LLMs and the human-expert reference) across the three OSCE cases (hematuria, fever in pregnancy, chronic cough). For each input prompt, the translated source-doctor question is followed by the five sources' translated responses. Korean originals are released alongside the static-evaluation materials. Speaker labels are abbreviated throughout this appendix: *Human* denotes the human-expert reference, and *Claude*, *Llama*, and *HyperCLOVA* denote Claude 3.5 Sonnet, Llama-3.1 70B, and HyperCLOVA X, respectively.

Case: Hematuria (red urine) *(translated from Korean)*

Q1. Hello, I'm medical student Kim Si-jun. May I have your name and age, please?

• Human: Hello, my name is Jung Jin-su, and I'm 28 years old.

• Claude: Hello, my name is Jung Jin-su, 28 years old.

• GPT-4o: Hello, I'm Jung Jin-su, 28 years old.

• HyperCLOVA: Jung Jin-su: Hello, my name is Jung Jin-su, 28 years old.

• Llama: Yes, my name is Jung Jin-su, 28 years old.

Q2. Hello, Mr. Jung. What brings you in today?

• Human: My urine suddenly turned red, so I came in.

• Claude: I came to the hospital because of red urine that started three days ago.

• GPT-4o: Hello, I'm here because red urine has been appearing for the past three days.

• HyperCLOVA: Yes. My urine has been red for three days, so I came in.

• Llama: I've had red urine since three days ago.

Q3. What was the color of the red urine like — closer to brown, or pinkish?

• Human: It seemed close to a dark brown.

• Claude: Initially it was deep reddish-brown, gradually lightening to about dark amber today.

• GPT-4o: At first it was deep reddish-brown, gradually lightening to about dark amber today.

• HyperCLOVA: On the first day it was very deep reddish-brown. Now it's lighter — about dark amber.

• Llama: Initially it was deep reddish-brown, gradually lightening to about dark amber today.

Q4. Was there any blood clot or stone in the urine?

• Human: No, there wasn't anything like that.

• Claude: I didn't see any clots or stones. There seemed to be a slight foam in the urine.

• GPT-4o: No, there were no clots or stones.

• HyperCLOVA: No, nothing like that. There seemed to be slight foam in the urine.

• Llama: No, there were no clots or stones.

Q5. Have you ever had red urine before? Was hematuria ever mentioned in school or workplace check-ups?

• Human: No, this is the first time. I've never been told about it on check-ups either.

• Claude: This is the first time I've had red urine. I've never been told about hematuria on check-ups.

• GPT-4o: No, I've never been told about hematuria, proteinuria, or kidney disease, and this is the first time the urine has appeared abnormal.

• HyperCLOVA: No, this kind of urine is the first time. I haven't heard about hematuria on check-ups either.

• Llama: No, I've never had red urine before, and the school and workplace check-ups never mentioned hematuria either.

Q6. That must have been distressing. Around the time the red urine appeared, did you have a cold or do strenuous exercise?

• Human: I had a cold for about a week. The red urine appeared while I was at the peak of being sick.

• Claude: Yes, I had cold symptoms before the red urine appeared, and my fever went up to 39 °C. I didn't exercise.

• GPT-4o: Yes, the day before the red urine I had a fever up to 39 °C with muscle aches and cold symptoms. I haven't exercised since the cold started a week ago.

• HyperCLOVA: Yes, I happened to have a cold, with severe fever and muscle aches in particular, so I took antipyretics.

• Llama: Yes, I had a cold. Cold symptoms started seven days ago, and the day before the red urine my fever rose to 39 °C with muscle aches.

Q7. When you had the red urine, did you have any flank pain or abdominal pain?

• Human: There was no abdominal or flank pain.

• Claude: No abdominal pain, flank pain, or pain at the urethral tip when I had red urine.

• GPT-4o: No, there was no abdominal pain, flank pain, or pain at the urethral tip with the red urine.

• HyperCLOVA: No, there was no pain at all when passing urine.

• Llama: No, there was no flank pain or abdominal pain.

Q8. Have you had any swelling of the legs or face?

• Human: No swelling.

• Claude: No swelling of the legs or face. My weight has not changed.

• GPT-4o: No, my weight hasn't changed and I haven't felt weak or had any swelling.

• HyperCLOVA: No weight change, and no particular swelling.

• Llama: No, I haven't had any swelling of the legs or face.

Q9. Have you had joint pain anywhere or red spots on the skin?

• Human: No, nothing like that.

• Claude: No joint pain or red spots on the skin.

• GPT-4o: No, there have been no muscle pain or red skin spots.

• HyperCLOVA: No, only the urine color was strange.

• Llama: No, I've had no joint pain or red skin spots.

Q10. When you passed red urine, did you have pain at the end of urination, or any discomfort?

• Human: No, there was no discomfort when urinating.

• Claude: I had no pain or discomfort at all when urinating.

• GPT-4o: No, no pain at the urethral tip or any other discomfort when urinating.

• HyperCLOVA: No, that wasn't the case. I was actually surprised at how normal it felt.

• Llama: No, no pain at the urethral tip or discomfort with urination.

Q11. Have you been urinating very frequently lately, or feeling that you can't fully empty your bladder?

• Human: No, I haven't had those symptoms.

• Claude: No increase in frequency or sense of incomplete emptying. No urinary discomfort.

• GPT-4o: No, no increased frequency or sense of incomplete emptying.

• HyperCLOVA: I don't drink much water, so I don't urinate often, and I haven't felt incomplete emptying.

• Llama: No, I haven't had increased frequency or incomplete emptying recently.

Q12. Have you noticed any foam in your urine?

• Human: Yes, I did notice some slight foam.

• Claude: I'm not sure, but there seemed to be a slight foam in the urine.

• GPT-4o: I'm not sure, but there seemed to be a slight foam in the urine.

• HyperCLOVA: Not sure, but I think there was a little.

• Llama: Yes, there seemed to be a slight foam in the urine.

Q13. Are you on any regular medication, or did you take anything for the cold recently?

• Human: When the cold was severe I bought Tylenol from the pharmacy and took it for a few days. I've stopped now.

• Claude: No regular medications. I bought Tylenol from the pharmacy for the cold.

• GPT-4o: No regular medication. I took Tylenol because of the cold symptoms.

• HyperCLOVA: I take no regular medications. I just bought and took Tylenol when I had the cold.

• Llama: Yes, I took Tylenol that I bought from the pharmacy for the cold.

Q14. Do you have any pre-existing kidney disease or have you had any major surgery?

• Human: No, nothing like that.

• Claude: No kidney disease or major surgery. I've considered myself healthy.

• GPT-4o: No, I've never been told about hematuria, proteinuria, or kidney disease, and I've never had major surgery.

• HyperCLOVA: No, I'm very healthy.

• Llama: No, no kidney disease or major surgery.

Q15. Does any family member have hematuria or kidney disease, or any urological cancer?

• Human: No, no family members with such illnesses, as far as I know.

• Claude: No family member has been diagnosed with hematuria, kidney disease, or urological cancer. My father has high blood pressure.

• GPT-4o: No, no family members have been diagnosed with hematuria, kidney disease, or urological cancer.

• HyperCLOVA: I know my father has high blood pressure and takes medication. Other than that, none.

• Llama: No, no family members have hematuria, kidney disease, or urological cancer.

Q16. Do you smoke?

• Human: Yes, about a pack a day.

• Claude: Yes, since my military service I've smoked about one pack a day.

• GPT-4o: Yes, I usually smoke about one pack a day.

• HyperCLOVA: Yes. I started in the military and smoke about one pack a day.

• Llama: Yes, since my military service I've smoked about one pack a day.

Q17. Do you drink alcohol? If so, how much?

• Human: I drink only at company gatherings, just a little.

• Claude: I only drink one or two glasses of beer at company gatherings.

• GPT-4o: Yes, I drink only at company gatherings, one or two glasses of beer.

• HyperCLOVA: Hardly. Just one or two glasses of beer at company gatherings.

• Llama: Yes, I drink one or two glasses of beer only at company gatherings.

Q18. I'd like to ask carefully — have you had any sexual contact with a stranger recently, or felt any pain after intercourse?

• Human: No, nothing like that.

• Claude: I only have sex with my girlfriend, about once a week. No contact with strangers and no pain.

• GPT-4o: No, no recent sexual contact with strangers, and no pain after intercourse.

• HyperCLOVA: I only have intercourse with my girlfriend, always with a condom. And nothing like that has happened.

• Llama: No, I haven't had any recent sexual contact with strangers, and no pain.

Q19. Is your urine stream weak, or do you have to strain to urinate?

• Human: No, neither.

• Claude: No, no weak stream and no straining required.

• GPT-4o: No, my urine stream isn't weak and I don't have to strain.

• HyperCLOVA: No, it's the same as usual.

• Llama: No, my stream isn't weak and I don't have to strain.

Q20. Thank you. To summarize: you had a cold a week ago, and at the peak of the fever you noticed reddish-brown urine, with no pain on urination, abdominal pain, or flank pain. Is that correct?

• Human: Yes, that's correct.

• Claude: Yes, that's right. After cold symptoms and high fever I had reddish-brown urine, with no pain.

• GPT-4o: Yes, that's right. A week ago I had cold symptoms; at the peak of fever I had reddish-brown urine and no pain or flank pain on urination.

• HyperCLOVA: Yes, exactly. That's what I said.

• Llama: Yes, that's correct.

Q21. Given the picture, this is likely post-cold-related hematuria. Hematuria means blood from the kidneys or urinary tract has mixed into the urine. Without pain or clots, stones are less likely; cold-related stress on the kidney glomerulus is a more probable cause. Does that make sense?

• Human: Yes, I understand. So what tests do I need?

• Claude: Yes, I understand. So you're saying cold-related stress likely affected the kidneys and caused hematuria.

• GPT-4o: Yes, I understand. The explanation about cold-related stress causing glomerular hematuria is clear. Thank you.

• HyperCLOVA: Yes, you mean it could be due to the cold, right?

• Llama: Yes, I understand.

Q22. For diagnosis we'll need a urinalysis, microscopy, proteinuria check, and a blood test for kidney function. If anything is abnormal we may need imaging or a kidney biopsy.

• Human: Alright.

• Claude: Yes, I understand. So urinalysis, blood tests, and kidney-function tests first, with additional tests if needed.

• GPT-4o: Yes, I understand. Urinalysis, microscopy, proteinuria testing, and blood-based kidney-function tests are needed; additional detailed tests if abnormal.

• HyperCLOVA: Yes, I see. Can I get the tests done right away?

• Llama: Yes, I understand.

Q23. Lastly, since hematuria may be associated with smoking, I would recommend quitting.

• Human: Alright. I'll quit smoking. Thank you.

• Claude: Yes, I'll consider quitting since smoking may be linked to hematuria.

• GPT-4o: Yes, I understand the link between smoking and hematuria, and I'll consider quitting. Thank you for the advice.

• HyperCLOVA: Yes, understood. I'll quit immediately.

• Llama: Yes, I understand. I'll quit smoking.

Case: Fever (in pregnancy) *(translated from Korean)*

Q1. Hello, I'm medical student Kang Min-woo. May I have your name and age, please?

• Human: Hello, I'm Lee Su-jin, 28 years old.

• Claude: Hello, my name is Lee Su-jin, 28 years old.

• GPT-4o: Hello, I'm Lee Su-jin, 28 years old.

• HyperCLOVA: Yes. Hello, I'm Lee Su-jin, 28 years old.

• Llama: Hello, I'm Lee Su-jin, 28 years old.

Q2. Hello, Ms. Lee. What brings you in today?

• Human: I came in because I had a high fever from yesterday evening.

• Claude: Since yesterday evening I've had a high fever and muscle pain all over. My right flank especially hurts a lot.

• GPT-4o: I came in because of fever since yesterday and pain in my right flank.

• HyperCLOVA: I came in because of fever and muscle pain all over since yesterday evening.

• Llama: I have severe body aches from the high fever, with sharp pain in the right flank in particular.

Q3. That sounds difficult. How high did the fever go, and is it constant or intermittent?

• Human: Yesterday evening it measured up to 39 °C, and even after Tylenol the fever barely came down.

• Claude: It went up to 39 °C on the thermometer. The fever was continuous, and Tylenol didn't bring it down well.

• GPT-4o: The fever went up to 39 °C, and it seems constant.

• HyperCLOVA: It went up to 39 °C and was sustained continuously.

• Llama: Measured at 39 °C. The fever started yesterday evening and has continued to now.

Q4. Did you have chills or shivering?

• Human: Yes, when it's bad I feel very cold and shivery.

• Claude: Yes, there were some chills. I felt shivery.

• GPT-4o: There were some chills.

• HyperCLOVA: Slight chills accompanied it.

• Llama: No, no cold sweats.

Q5. Any sore throat, cough, or sputum?

• Human: No sore throat, cough, or sputum.

• Claude: No, no sore throat, cough, or sputum.

• GPT-4o: No, no cough or sputum.

• HyperCLOVA: No cough, no sputum.

• Llama: No, no cough. No sputum either.

Q6. Any diarrhoea or vomiting?

• Human: No.

• Claude: No vomiting and no diarrhoea, though I did feel a bit nauseous.

• GPT-4o: No, no diarrhoea or vomiting.

• HyperCLOVA: No diarrhoea, no vomiting.

• Llama: No, no vomiting. No diarrhoea either.

Q7. Any pain in the flank or lower abdomen?

• Human: Yes, the right flank in particular hurts.

• Claude: Yes, sharp pain in the right flank when moving. The lower abdomen isn't particularly painful.

• GPT-4o: Yes, sharp pain in the right flank.

• HyperCLOVA: Sharp pain in the right flank when moving.

• Llama: No, the flank pain has eased somewhat, and no lower-abdominal pain.

Q8. What is the right flank pain like? Please describe in detail.

• Human: It's fine when still, but when I try to move, the right flank stabs sharply.

• Claude: The right flank pain is worse on movement, sharp and stabbing. After Tylenol it improved a bit but is still uncomfortable.

• GPT-4o: Sharp, stabbing pain on movement.

• HyperCLOVA: Pain when changing posture; even at rest there's a throbbing sensation.

• Llama: Sharp pain in the right flank on movement.

Q9. Was the urine color normal? Any red tint or clots?

• Human: It was fine until yesterday, but this morning the urine came out a bit red.

• Claude: This morning when I urinated there was a slight red blood tinge. It's a new symptom and I'm worried.

• GPT-4o: This morning a slight red blood tinge appeared in the urine.

• HyperCLOVA: This morning when I urinated a slight red blood tinge appeared.

• Llama: No, a slight red blood tinge appeared in the urine.

Q10. Any pain or discomfort when urinating?

• Human: Now that you mention it, I have been urinating frequently lately.

• Claude: Yes, for the past week I've been urinating frequently with some tightness and discomfort. I thought it was from the pregnancy.

• GPT-4o: There was some tightness and discomfort when urinating.

• HyperCLOVA: For the past week I've been urinating frequently with some tightness and discomfort.

• Llama: No, but I had felt some tightness and discomfort when urinating.

Q11. Any signs of inflammation or bleeding from the vaginal area?

• Human: No, none.

• Claude: No, no inflammation or bleeding from the vaginal area.

• GPT-4o: No, no inflammation or bleeding from the vaginal area.

• HyperCLOVA: No, none.

• Llama: No, no vaginal inflammation or bleeding.

Q12. Are you pregnant?

• Human: Yes, I am 20 weeks pregnant.

• Claude: Yes, I'm 20 weeks pregnant. So far the pregnancy has been progressing well.

• GPT-4o: Yes, I'm 20 weeks pregnant.

• HyperCLOVA: Yes, 20 weeks pregnant.

• Llama: Yes, I'm 20 weeks pregnant.

Q13. Is this your first pregnancy? Have you had previous pregnancies?

• Human: Yes, this is my first.

• Claude: This is my first pregnancy. I've never had a miscarriage or termination.

• GPT-4o: Yes, this is my first pregnancy. I've had no previous pregnancies, miscarriages, or terminations.

• HyperCLOVA: Yes, this is my first.

• Llama: Yes, first pregnancy. No previous pregnancies.

Q14. Are you on any regular medications?

• Human: No, just Tylenol that I bought yesterday for the fever.

• Claude: No regular medications. I just took Tylenol yesterday for the fever.

• GPT-4o: No, no regular medications.

• HyperCLOVA: Just Tylenol I bought yesterday; no other regular medications.

• Llama: No, no regular medications.

Q15. Have you had any major illness in the past, or been diagnosed with diabetes?

• Human: No, nothing in particular.

• Claude: No history of major illness or diabetes diagnosis. I haven't had abnormal findings on antenatal exams.

• GPT-4o: No major illness. No diabetes diagnosis.

• HyperCLOVA: No major illness. No diabetes either.

• Llama: No major illness, no diabetes diagnosis.

Q16. Have you ever had cystitis or a urinary-tract infection before?

• Human: No, never.

• Claude: No, no history of cystitis or urinary-tract infection.

• GPT-4o: No.

• HyperCLOVA: No.

• Llama: No, never.

Q17. Have you been drinking less water or eating less recently?

• Human: Yes, since pregnancy I drink frequently, so I've been deliberately drinking a bit less.

• Claude: I've been drinking a bit less water lately because I was urinating so often, but my appetite is normal.

• GPT-4o: I've been drinking less water lately because of the frequent urination.

• HyperCLOVA: Yes, I've been drinking less water because of frequent urination.

• Llama: Yes, I've been drinking less water lately because of frequent urination.

Q18. Any recent international travel or contact with someone who had infectious symptoms?

• Human: No.

• Claude: No international travel or contact with sick people.

• GPT-4o: No.

• HyperCLOVA: No.

• Llama: No, no recent travel or contact with infectious patients.

Q19. Do you drink alcohol or smoke?

• Human: I used to drink occasionally, but quit since pregnancy.

• Claude: I drink only one beer occasionally and never since pregnancy. I don't smoke.

• GPT-4o: No alcohol or smoking since pregnancy.

• HyperCLOVA: I rarely drank — just one beer — and have quit since pregnancy. I don't smoke.

• Llama: No alcohol since pregnancy. No smoking.

Q20. Have you had sexual relations with your husband recently?

• Human: Yes, I've started carefully a few weeks ago after my obstetrician said it was safe.

• Claude: Yes, after my obstetrician said it was safe a few weeks ago we resumed carefully.

• GPT-4o: Yes, we've resumed carefully a few weeks ago after the obstetrician's advice.

• HyperCLOVA: Yes, we resumed carefully a few weeks ago.

• Llama: Yes, we resumed carefully after the obstetrician advised it was safe.

Q21. Thank you. Based on the symptoms, your fever is most likely from acute pyelonephritis. This is a kidney infection, often starting as cystitis and ascending the ureter. Does that make sense?

• Human: Yes, I understand. So what should I do?

• Claude: Yes, I see. So a urinary-tract infection that ascended to the kidney has caused the fever and flank pain.

• GPT-4o: Yes, I understand. So a kidney infection causes the fever and flank pain. Thank you.

• HyperCLOVA: Yes. Cystitis ascending to the kidney caused the symptoms, then?

• Llama: Yes, I understand.

Q22. For a pregnant patient with acute pyelonephritis like yours, antibiotic treatment is the principle for both maternal and fetal safety. You should also drink about 2 L of water daily, not hold urine, and take antibiotics regularly. Any questions?

• Human: Wait — is the antibiotic safe for the baby?

• Claude: I'm worried whether antibiotics will be harmful to the baby. Are pregnancy-safe medications available?

• GPT-4o: Yes, I understand. But I'm worried about the safety of antibiotics for the baby.

• HyperCLOVA: Will antibiotics affect the baby?

• Llama: Yes. But I'm worried about whether antibiotics may harm the baby.

Q23. All medications carry some risk, but if pyelonephritis is left untreated the inflammation can endanger the baby more. We start treatment early with the safest available antibiotics — risk varies by drug, and we'll choose ones with the lowest known risk for fetal abnormalities and safety concerns.

• Human: Alright. I'll trust your judgement.

• Claude: I understand. So we'll use safer antibiotics whose pregnancy-related risks have been assessed. Thank you for the explanation.

• GPT-4o: Yes, I understand. So untreated infection is more dangerous than antibiotic risk, and you'll use safer options.

• HyperCLOVA: I see. So treating the infection is more important than not treating it.

• Llama: Yes, I understand.

Case: Chronic cough *(translated from Korean)*

Q1. Hello, I'm medical student Cho Su-jin. May I have your name and age, please?

• Human: Hello, I'm Lee Yun-jung, 55 years old.

• Claude: Hello, doctor. My name is Lee Yun-jung and I'm 55 years old.

• GPT-4o: Hello, I'm Lee Yun-jung, 55 years old.

• HyperCLOVA: My name is Lee Yun-jung, 55 years old.

• Llama: Hello! My name is Lee Yun-jung, 55 years old.

Q2. Hello, Ms. Lee. What brings you in today?

• Human: My cough won't stop. It's been more than three months, so I came in.

• Claude: I came in because of a cough that's been ongoing for three months. I thought it was nothing at first, but it gradually got worse.

• GPT-4o: I have a cough that's been continuing for three months. It's mostly in the morning, and it's a dry cough.

• HyperCLOVA: I came in because of a cough lasting three months. At first I thought it was a cold, but it's gone on so long that I'm worried.

• Llama: Lately I've had a cough for three months. It's worse in the morning and is a dry cough. There's almost no sputum, but sometimes I feel something stuck in my throat.

Q3. That sounds difficult. Let me ask a few questions to help diagnose. When during the day do you cough most?

• Human: Mostly in the early morning or just after dawn.

• Claude: I cough mostly in the early morning. It tends to be a bit more severe when I wake up.

• GPT-4o: Mostly early in the morning.

• HyperCLOVA: Mostly in the morning.

• Llama: Mostly in the morning, especially early. Then around lunchtime, and a bit less in the evening.

Q4. Are there particular situations where the cough comes — at night, after eating, or other?

• Human: When I eat spicy food or overeat, the cough seems worse.

• Claude: Now that I think about it, after meals when my chest feels heavy and burning the cough is worse. Spicy or oily foods make it worse, and at night too.

• GPT-4o: After meals when my chest feels heavy and burning the cough comes. It's worse at night.

• HyperCLOVA: I do cough at night, but more after meals — especially with oily or spicy food.

• Llama: After meals, especially after oily or spicy food. Also at night, when stomach acid feels like it's coming up.

Q5. Do you have sputum or runny nose with the cough?

• Human: No sputum or runny nose.

• Claude: Sputum is rare. It's mostly a dry cough. Sometimes I feel something stuck in my throat. No runny nose.

• GPT-4o: No sputum or runny nose. Sometimes I feel something stuck in my throat.

• HyperCLOVA: Sputum is rare; sometimes I feel something stuck in my throat. No runny nose.

• Llama: No, sputum is rare. No runny nose either. Just the cough.

Q6. Have you had fever or chills?

• Human: No, no fever or cold symptoms — just the cough.

• Claude: No fever, chills, or body aches with the cough.

• GPT-4o: No fever or chills.

• HyperCLOVA: Nothing like that. Just the cough.

• Llama: No, no fever, chills, or body aches with the cough.

Q7. Any post-nasal drip — feeling that mucus is dripping down the back of your throat?

• Human: No, that doesn't happen.

• Claude: Hmm:. I haven't noticed that. No particular sensation of mucus dripping.

• GPT-4o: No, no such sensation.

• HyperCLOVA: No.

• Llama: No, no post-nasal drip. Sometimes I feel something stuck in my throat, but only occasionally.

Q8. Any vomiting, diarrhoea, or constipation?

• Human: No.

• Claude: No vomiting, diarrhoea, or constipation. My bowel habits are unchanged.

• GPT-4o: No, no such symptoms.

• HyperCLOVA: No, but I've had a burning sensation in my chest, especially after meals.

• Llama: No, no vomiting, diarrhoea, or constipation.

Q9. Does the cough get worse when you lie down?

• Human: It gets worse on waking, so when lying down it tends to come more.

• Claude: Yes, the cough is worse when lying down. Especially at night when I try to sleep.

• GPT-4o: Yes, it seems worse when lying down.

• HyperCLOVA: Lying down doesn't particularly make it worse.

• Llama: No, the cough comes when sitting or standing. Lying down too, but mostly worse at night after oily or spicy food.

Q10. Have you had any swelling or shortness of breath?

• Human: No.

• Claude: No swelling or shortness of breath. I'm living normally with no shortness of breath on exercise.

• GPT-4o: No swelling or shortness of breath.

• HyperCLOVA: No swelling, no shortness of breath.

• Llama: No, no swelling or shortness of breath. No chest pain.

Q11. What is your height and weight?

• Human: I'm 160 cm and 54 kg.

• Claude: Height 160 cm, weight 54 kg.

• GPT-4o: 160 cm, 54 kg.

• HyperCLOVA: 160 cm, 54 kg.

• Llama: 160 cm, 54 kg.

Q12. Has your weight changed compared to six months ago?

• Human: No, no significant change.

• Claude: No, my weight hasn't really changed over the past six months.

• GPT-4o: No.

• HyperCLOVA: No.

• Llama: No, no weight change.

Q13. Any burning sensation in the chest, or stomach acid coming up?

• Human: Yes, especially after spicy food my chest feels burning.

• Claude: Yes, after meals — especially spicy ones — my chest feels burning, and sometimes I feel acid rising into my throat.

• GPT-4o: Yes, particularly after spicy food my chest feels burning.

• HyperCLOVA: Yes, my chest feels burning after meals, and sometimes I feel acid rising.

• Llama: Yes, after spicy food my chest feels burning, and acid rises into my throat.

Q14. Are there triggers like cold air, dust, or allergens that make the cough worse?

• Human: Not particularly.

• Claude: Not particularly. Cold air, dust, or pollen doesn't seem to make it worse.

• GPT-4o: No, no specific environmental triggers.

• HyperCLOVA: No.

• Llama: No, no environmental triggers like cold air, dust, or pollen.

Q15. Does the cough come and go with the seasons, with runny nose at certain times?

• Human: No, it's not seasonal.

• Claude: No, it's not particularly seasonal. No previous diagnosis of allergic rhinitis.

• GPT-4o: No, no seasonal pattern.

• HyperCLOVA: No, no seasonal pattern.

• Llama: No, no seasonal cough or runny nose.

Q16. Do you have any pre-existing lung conditions, allergic rhinitis, or GERD diagnosis?

• Human: No, I've never had a lung disease or rhinitis diagnosis.

• Claude: No previous diagnosis of lung disease, allergic rhinitis, or GERD.

• GPT-4o: No, no such diagnoses.

• HyperCLOVA: No.

• Llama: No, no diagnoses of lung disease, allergic rhinitis, or GERD.

Q17. Are you on any regular medications?

• Human: I take amlodipine for high blood pressure for the past 5 years.

• Claude: Yes, amlodipine for hypertension, for about five years. The blood pressure is well controlled.

• GPT-4o: Yes, I take amlodipine daily for hypertension.

• HyperCLOVA: Yes, amlodipine for blood pressure.

• Llama: Yes, amlodipine for hypertension.

Q18. Does anyone in your family have lung cancer, or any major lung or bronchial disease?

• Human: No, no family members with lung disease.

• Claude: No family member has lung cancer or major lung disease. My mother has the same hypertension and takes medication.

• GPT-4o: No, no family members with lung disease.

• HyperCLOVA: No.

• Llama: No, no family members with lung disease.

Q19. Do you smoke or drink alcohol?

• Human: I've never smoked. Just a little drink at gatherings.

• Claude: I've never smoked. Drink occasionally — only at company gatherings.

• GPT-4o: No smoking. Occasional drinking at gatherings.

• HyperCLOVA: No smoking. Drink occasionally.

• Llama: I've never smoked. I drink only at company gatherings.

Q20. What is your occupation? Are there toxic substances or dust in your work environment?

• Human: I work in office management at a supermarket. I'm mostly in the office.

• Claude: I work in office administration at a mart. The work environment is normal — no toxic substances or dust.

• GPT-4o: I work in office administration at a mart. No toxic substance or dust exposure.

• HyperCLOVA: I work at a mart in office administration; the environment isn't dusty.

• Llama: I work in office administration at a mart, mostly in the office. No toxic or dusty environment.

Q21. Have you had recent contact with someone with severe fever or cough?

• Human: No, no recent contacts with sick people.

• Claude: No, no recent contact with people who had severe cold or fever.

• GPT-4o: No.

• HyperCLOVA: No.

• Llama: No, no recent contact with infectious patients.

Q22. Thank you. Based on what you've described, the most likely cause of your cough is GERD-related cough. Even without pneumonia or bronchitis, when stomach acid refluxes, it can trigger cough. Does that make sense?

• Human: Yes, I understand. So GERD is causing the cough?

• Claude: Yes, I understand. So GERD is causing the cough, not lung disease.

• GPT-4o: Yes, I understand. So acid reflux is the suspected cause of the cough.

• HyperCLOVA: Yes, GERD-related cough.

• Llama: Yes, I understand. So GERD is causing the chronic cough.

Q23. For diagnosis, an endoscopy to check the oesophagus and possibly a 24-hour acid-monitoring probe may be needed.

• Human: Alright, I understand.

• Claude: Yes, I understand. Endoscopy and possibly 24-hour acid monitoring.

• GPT-4o: Yes, I understand.

• HyperCLOVA: Yes, I understand.

• Llama: Yes, I understand. Endoscopy and possibly acid monitoring.

Q24. And since GERD is suspected, for prevention and improvement: sleep with a slightly elevated pillow, avoid overeating and irregular meals, and try to avoid coffee, soft drinks, and juices.

• Human: Alright, I'll try.

• Claude: Yes, I'll try the elevated pillow, regular meals, and avoid coffee and soft drinks.

• GPT-4o: Yes, I'll try those lifestyle changes.

• HyperCLOVA: Yes, I'll do that.

• Llama: Yes, I'll try those lifestyle changes.

D Per-Case Encounter-Level Likert Scores

This appendix provides per-case mean Likert scores with 95% confidence intervals for the six encounter-level metrics (Comprehensibility, Hallucination-free, Coherence, Consistency, Engagement, Overall Satisfaction), broken down across the three OSCE cases. The pooled-across-cases values are reported in Table 3 of the main paper. Per-case radar plots (Figures D1–D3) visualize the same data to make per-case profile differences across the five sources directly comparable.

Table D1: Per-case mean encounter-level Likert score [95% bootstrap CI] for each source × metric, broken down by OSCE case (n = 17 ratings per cell).

Metric	Human (reference)	Claude 3.5 Sonnet	GPT-4o	Llama-3.1 70B	HyperCLOVA X
Hematuria					
Comprehensibility	4.82 [4.65–5.00]	4.53 [4.24–4.82]	4.41 [3.94–4.76]	4.53 [4.24–4.82]	4.71 [4.47–4.88]
Hallucination-free	4.76 [4.53–4.94]	4.24 [3.76–4.65]	4.29 [3.94–4.65]	4.59 [4.35–4.82]	4.41 [4.12–4.71]
Coherence	4.71 [4.41–4.94]	3.71 [3.35–4.12]	3.41 [3.00–3.82]	3.88 [3.41–4.35]	4.35 [4.00–4.65]
Consistency	4.71 [4.47–4.88]	4.41 [4.00–4.76]	4.53 [4.24–4.82]	4.53 [4.24–4.82]	4.59 [4.29–4.88]
Engagement	3.71 [3.12–4.24]	3.53 [3.12–4.00]	3.35 [2.94–3.82]	3.71 [3.23–4.18]	4.35 [3.94–4.71]
Overall Satisfaction	4.59 [4.24–4.88]	3.59 [3.06–4.06]	3.18 [2.71–3.71]	3.94 [3.47–4.41]	4.12 [3.82–4.41]
Fever (in pregnancy)					
Comprehensibility	4.88 [4.71–5.00]	4.65 [4.35–4.88]	4.76 [4.53–4.94]	3.18 [2.71–3.65]	4.41 [4.12–4.71]
Hallucination-free	4.65 [4.41–4.88]	4.59 [4.29–4.88]	4.59 [4.29–4.88]	3.71 [3.18–4.18]	4.12 [3.76–4.47]
Coherence	4.94 [4.82–5.00]	4.41 [4.06–4.71]	4.41 [4.12–4.71]	2.59 [2.24–2.94]	4.06 [3.65–4.41]
Consistency	4.88 [4.71–5.00]	4.76 [4.53–4.94]	4.65 [4.41–4.88]	3.53 [3.00–4.12]	4.41 [4.18–4.65]
Engagement	4.24 [3.94–4.53]	4.00 [3.59–4.35]	3.76 [3.35–4.18]	2.53 [2.12–3.00]	3.88 [3.47–4.29]
Overall Satisfaction	4.76 [4.47–5.00]	4.24 [3.71–4.65]	4.12 [3.82–4.41]	2.24 [1.82–2.76]	3.71 [3.29–4.06]
Chronic cough					
Comprehensibility	4.82 [4.65–5.00]	4.18 [3.82–4.53]	4.59 [4.24–4.88]	3.41 [3.06–3.82]	4.59 [4.29–4.88]
Hallucination-free	4.06 [3.65–4.41]	4.24 [3.88–4.59]	4.35 [3.94–4.71]	3.53 [3.12–3.94]	3.53 [3.06–4.00]
Coherence	4.76 [4.53–4.94]	3.88 [3.35–4.35]	4.41 [4.06–4.71]	2.35 [2.00–2.71]	4.29 [3.94–4.65]
Consistency	4.65 [4.24–4.94]	4.00 [3.53–4.47]	4.41 [4.06–4.76]	3.41 [2.94–3.88]	4.18 [3.82–4.53]
Engagement	4.59 [4.29–4.82]	3.94 [3.41–4.47]	4.12 [3.76–4.47]	2.59 [2.06–3.12]	4.24 [3.88–4.59]
Overall Satisfaction	4.47 [4.12–4.76]	3.65 [3.12–4.18]	4.18 [3.76–4.53]	1.94 [1.59–2.35]	3.82 [3.35–4.29]

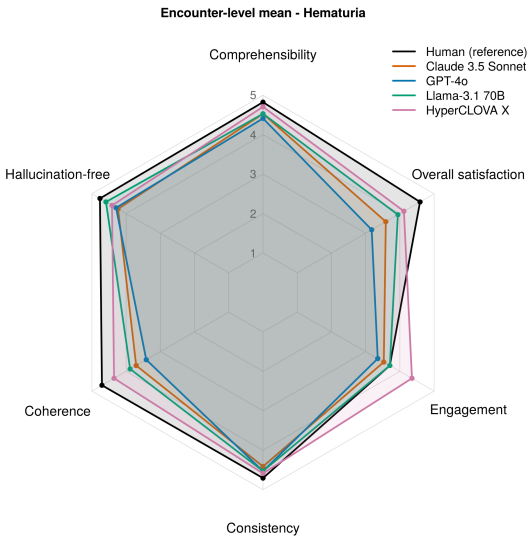


Figure D1: Per-case encounter-level Likert profile — Hematuria. Each axis is one of the six encounter-level metrics; values plotted are mean Likert scores (n = 17 raters). Five overlaid polygons compare Human (reference), Claude 3.5 Sonnet, GPT-4o, Llama-3.1 70B, and HyperCLOVA X.

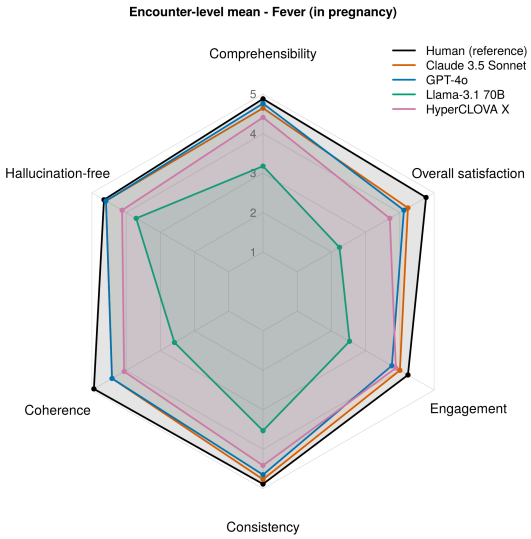


Figure D2: Per-case encounter-level Likert profile — Fever in pregnancy. Axes and overlay sources are the same as in Figure D1.

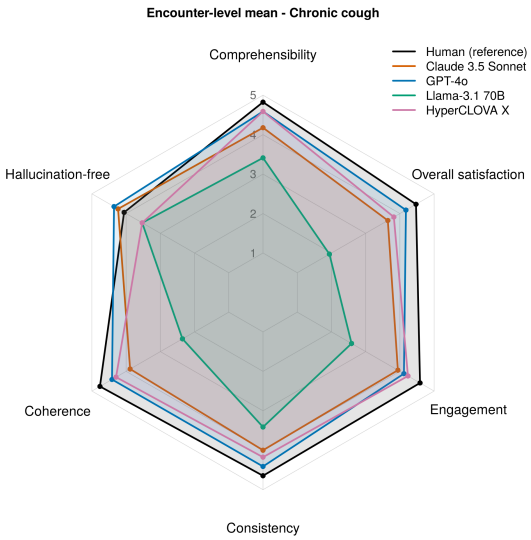


Figure D3: Per-case encounter-level Likert profile — Chronic cough. Axes and overlay sources are the same as in Figure D1.

E Inter-Rater Reliability and Linguistic-Form Means

This appendix provides the full inter-rater reliability statistics referenced in §5.4 and the Source $\times$ Linguistic-form interaction figure referenced in §5.3.

Table E1: Inter-rater reliability across 17 medical-student raters. ICC(2,k) is reported with 95% confidence intervals (F-distribution method); Krippendorff's α uses ordinal weights.

Level	**Metric / Source**	$n_{\textbf{targets}}$	**ICC(2,k) [95% CI]**	**Krippendorff α**
Encounter	Comprehensibility	15	0.91 [0.82–0.96]	0.26
Encounter	Hallucination-free	15	0.79 [0.61–0.91]	0.14
Encounter	Coherence	15	0.94 [0.89–0.98]	0.41
Encounter	Consistency	15	0.83 [0.68–0.93]	0.15
Encounter	Engagement	15	0.86 [0.73–0.94]	0.22
Encounter	Overall Satisfaction	15	0.93 [0.86–0.97]	0.35
Encounter	**Fully pooled**	90	**0.91 [0.88–0.94]**	**0.30**
Sentence	Human (reference)	210	0.80 [0.75–0.84]	0.08
Sentence	Claude 3.5 Sonnet	210	0.81 [0.77–0.85]	0.18
Sentence	GPT-4o	210	0.82 [0.78–0.86]	0.15
Sentence	Llama-3.1 70B	210	0.92 [0.90–0.94]	0.33
Sentence	HyperCLOVA X	210	0.80 [0.76–0.84]	0.11

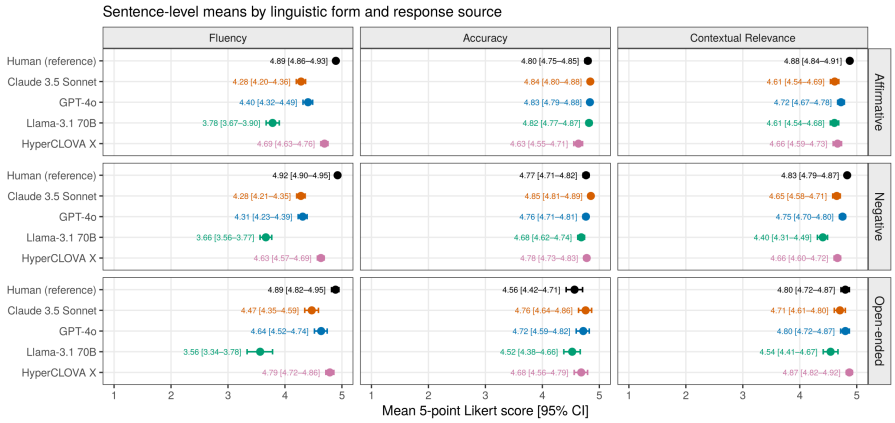


Figure E1: Sentence-level mean Likert score by linguistic form and response source. Panel rows are question forms (Affirmative, Negative-polarity, Open-ended), and panel columns are the three sentence-level metrics; within each panel the y-axis lists the five response sources, and the x-axis is the mean score. Points are means with 95% bootstrap confidence intervals.

F Platform Corpus Category Distribution

This appendix reports the per-category distribution referenced in §6.2.

Table F1: Distribution of the eleven history-taking categories in the 70-prompt evaluation set and in the platform-collected corpus (49 sessions, 27 scenarios, 1,763 utterances). Values are n (%); platform percentages are of the 1,614 utterances the scheme covers. Non-question directives and Rapport-building talk types fall outside the scheme and are reported as counts only. Platform utterances were labeled with the same rule-based tagger as the prompt set and corrected in full by one researcher.

Category	**Prompt set**	**Platform**
Greeting & ID check	3 (4.3)	62 (3.8)
Chief complaint (C.C.)	3 (4.3)	29 (1.8)
HPC — Character	10 (14.3)	302 (18.7)
HPC — Associated symptoms	21 (30.0)	463 (28.7)
HPC — Aggravating/relieving	3 (4.3)	106 (6.6)
Past medical history	6 (8.6)	200 (12.4)
Drug history	3 (4.3)	75 (4.6)
Family history	2 (2.9)	68 (4.2)
Social history	9 (12.9)	157 (9.7)
Clinical reasoning	2 (2.9)	58 (3.6)
Patient education	8 (11.4)	94 (5.8)
Total within the scheme	**70**	**1,614**
Outside the scheme		
Non-question directives	—	67
Rapport-building talk	—	82